



Original Article

Simulation of shockwave propagation characteristics in nuclear reactor cavity during external steam explosion using unified SPH

JinHyun Kim, Hee Sang Yoo, Young Beom Jo, Eung Soo Kim^{*}

Department of Nuclear Engineering, Seoul National University, Seoul, South Korea

ARTICLE INFO

Keywords:

Steam explosion
Unified Smoothed Particle Hydrodynamics (SPH)
GPU acceleration
APR1400 reactor cavity
Shockwave propagation
Structural Impact

ABSTRACT

Steam explosions in nuclear reactors pose significant risks to reactor safety and containment integrity during severe accidents. This study addresses the challenges of accurately simulating shockwave propagation and structural impact in such events by establishing a unified Smoothed Particle Hydrodynamics (SPH) framework. The proposed SPH model was optimized using GPU parallelization and validated against experimental results from shock tube, underwater explosion and high-velocity impact tests, demonstrating its efficacy in capturing shockwave-structure interactions. The model was applied to simulate steam explosions in the APR1400 reactor cavity under various conditions, examining factors such as the equation of state (EOS) for water, structural materials and explosive forces. The study further found that the choice of EOS and structural material greatly influenced the peak pressure and the extent of damage, with the Mie-Grüneisen EOS producing higher peak pressures and reinforced concrete showing higher damage resistance. These findings suggest that the unified SPH approach provides a comprehensive and robust framework for assessing steam explosion risks, offering crucial insights for optimizing reactor design and safety measures against such events.

1. Introduction

Steam explosions in the nuclear field are critical due to their potential to significantly impact reactor safety and containment integrity during severe accidents. When the molten corium interacts with coolant water, the intense and rapid heat transfer can lead to a steam explosion, resulting in high-pressure shock waves and potential structural damage [1]. Such explosions pose a serious risk as they can compromise the primary containment systems of nuclear reactors, potentially leading to the release of radioactive materials into the environment [2]. Therefore, understanding and mitigating the potential structural damage from steam explosions is crucial for ensuring the safety of nuclear power plants.

Despite the critical nature of this issue, there is limited research on the specific loads generated by explosions and their subsequent effects on reactor components or structures. This is due to the complexity of steam explosion phenomena, which involve intricate interactions between thermal, fluid, and mechanical processes [3]. Additionally, steam explosion experiments have been conducted only under limited conditions, as they require highly controlled environments and specialized equipment to replicate the extreme conditions of a nuclear reactor [4,5].

Previous approaches to addressing this issue have primarily focused on coupling explosion simulations with structural analysis. For instance, steam explosion calculations have been conducted using Computational Fluid Dynamics (CFD) or hydrodynamic codes to obtain transient pressure loads. These pressure loads are then used in structural analyses performed by Finite Element Method (FEM) codes to assess potential damage. A CFD code, CFX, was used to calculate the pressure, velocity, and volume fraction of the flow from the pressure wave of the multi-phase flow during steam explosion, and a structural analysis code, ABAQUS, was used to calculate the load on the cavity wall [1]. The pressure history by height was obtained with the TEXAS-V code under various conditions of corium condition, failure mode, explosion location, and turbulence model, and the stress and displacement of the reactor cavity and penetration piping were analyzed using the structural code ANSYS civil FEM [6]. For the most severe case from Ref. [6], various concrete models were used to examine the displacement and crushing region crack region according to the material model [7]. The pressure history of steam explosion after IVR-ERVC state was obtained with TRACER-II code, and the stress and strain of concrete wall and rebar were analyzed with ANSYS [8].

However, they involve complex and disjointed code coupling

^{*} Corresponding author. Department of Nuclear Engineering Seoul National University, Seoul, South Korea.

E-mail address: kes7741@snu.ac.kr (E.S. Kim).

<https://doi.org/10.1016/j.net.2024.10.036>

Received 30 August 2024; Accepted 19 October 2024

Available online 23 October 2024

1738-5733/© 2024 Korean Nuclear Society, Published by Elsevier Korea LLC. All rights are reserved, including those for text and data mining, AI training, and similar technologies. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

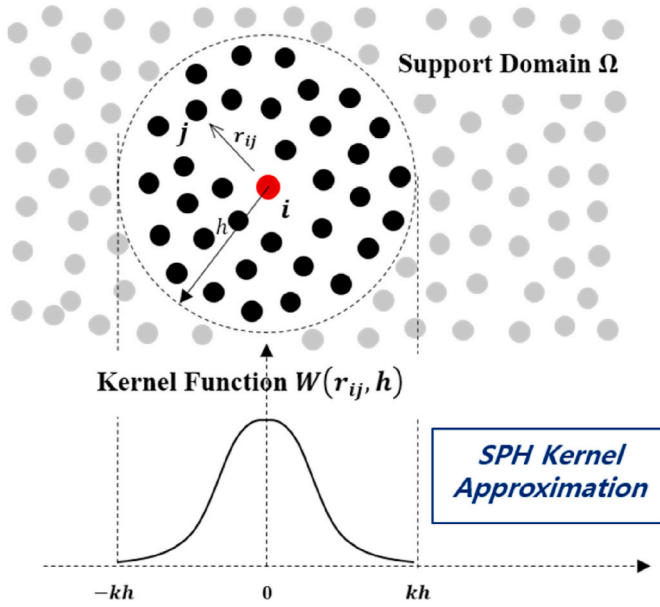


Fig. 1. SPH approximation and kernel function.

processes, resulting in significant challenges such as handling disparate material deformations and ensuring accurate mesh resolution [9]. Consequently, a few researches on the unified modeling framework have been conducted. The explosive load of the vapor explosion caused by FCI with TNT was simulated using the Coupled Eulerian Lagrangian (CEL) technique, and the Von Mises stress, crushing region, and crack region of various concrete material models were examined using ABAQUS [10]. The propagation of the shock wave from the explosion and the resulting loads on the structure were analyzed using the Arbitrary Lagrangian Eulerian (ALE) and Fluid Structure Interaction (FSI) methods, utilizing LS-DYNA [11]. These studies indicated that although some progress has been made, the unified modeling of these phenomena remains difficult due to the complexity of shockwave behavior, geometries, and mechanical interactions.

Smoothed Particle Hydrodynamics (SPH), a representative particle-based method, can address these challenges by focusing on the unified simulation of shockwave propagation and structural analysis due to its inherent advantages in managing complex nonlinearities. Shockwave propagation is a critical factor in understanding structural damage, and SPH offers significant advantages for this type of simulation. Unlike traditional grid-based approaches, SPH does not rely on a predefined mesh, making it inherently well-suited for handling fluid-structure interactions and managing large deformations, such as explosions and fractures. This mesh-free method eliminates issues related to mesh distortion and remeshing, thereby enhancing the stability of numerical analysis and feasibility for complex numerical simulations. Although SPH and other particle-based methods were initially less favored due to lower computational efficiency compared to grid-based methods, advancements in computing technology, especially the development of Graphics Processing Units (GPUs), have significantly improved their computational performance, making them increasingly viable for developing a unified calculation framework that can seamlessly integrate fluid and structural analysis.

In this context, this study established a unified SPH framework and applied it to the analysis of steam explosions and cavity assessments within nuclear reactors. Specifically, the SPH method was enhanced by incorporating a variable smoothing length, an integrated mass conservation equation, and suitable equations of state and constitutive models for various materials, ensuring robustness, accuracy, and reliability across different scenarios. The method's performance was further optimized through GPU parallelization, allowing for efficient management

of a large number of particles and improved computational efficiency. The phenomena of an explosion in a nuclear reactor cavity were then simulated, comparing various conditions such as structural analysis, EOS, reinforcement ratio, and explosive force, while measuring shock wave propagation and structural integrity. This approach provides a more comprehensive understanding of explosion dynamics and their impact on reactor structures. Ultimately, this research is expected to contribute significantly to the field by offering a robust and unified approach to modeling these complex interactions, enhancing the safety of nuclear reactors against steam explosion events.

This paper is organized as follows: Section 2 offers an overview of the SPH method and details the various models utilized in this research. Section 3 addresses the numerical validation of shock wave propagation and material models through problems involving shock waves, high-velocity impacts, and underwater explosions. Section 4 presents simulations of explosions in a reactor cavity, including comparisons under various conditions and analyses from the perspectives of shock wave propagation and structural integrity. Finally, Section 5 concludes the paper by summarizing the findings and offering recommendations for future research in this field.

2. Methodology

2.1. Fundamentals of SPH

Smoothed Particle Hydrodynamics (SPH) is a computational method for simulating fluid flows and solid mechanics problems. The method involves discretizing a continuous fluid or solid into individual particles, which interact with each other through pairwise forces. Each particle is assigned a number of properties, such as density, pressure, and velocity, which can be updated over time as the simulation progresses. One of the key advantages of SPH is its ability to handle complex geometries and free surfaces. It is particularly well-suited to explosion and impact problems because it can model the shock wave propagation, material melting, vaporization, and other complex physics associated with these extreme impact phenomena. Additionally, SPH's adaptability to parallel computing on graphical processing units (GPUs) enhances its capability to provide accurate and detailed predictions in solid material interaction problems.

The SPH method relies on representing a function $f(x)$ through an integral, which is utilized in the SPH method, originates from the subsequent identity.

$$f(x) = \int_{\Omega} \delta(x - x') f(x') dx' \quad (1)$$

Where f is a function of position vector x and Ω is the volume of integral which contains x . $\delta(x - x')$ is the Dirac delta function which is zero everywhere except at zero, where it is undefined, but has the property of integrating to one over the region.

For real world computation, the Dirac-delta function is replaced by kernel function $W(x - x', h)$ in SPH method.

$$\langle f(x) \rangle \simeq \int_{\Omega} W(x - x', h) f(x') dx' \quad (2)$$

To calculate numerical approximated integral formation in computer, discretized formulation is used.

$$f_i(r) = \sum_j \frac{m_j}{\rho_j} f_j W(r_i - r_j, h_i) \quad (3)$$

Here i indicates center particle, j indicates neighbor particle which used in SPH approximation and m is mass, ρ is density. Fig. 1 shows the concept of SPH approximation and kernel function.

Table 1
SPH formulation for governing equation.

Governing equation	SPH formulation
Mass conservation	$\rho_i = \sum_j \frac{\rho_{0j}}{\rho_{0i}} m_j W(h_i)$ (4)
	$\frac{d\rho_i}{dt} = \frac{1}{\Omega_i} \sum_j m_j (\mathbf{v}_i - \mathbf{v}_j) \cdot \nabla_i W_{ij}(h_i)$ (5)
Momentum conservation	$\frac{d\mathbf{v}_i}{dt} = \sum_j m_j \left[\frac{\sigma_j}{\Omega_j \rho_j^2} \nabla_i W_{ij}(h_i) + \frac{\sigma_j}{\Omega_j \rho_j^2} \nabla_i W_{ij}(h_j) - \Pi_{ij} \bullet \nabla_i W_{ij} \right]$ (6)
Energy conservation	$\frac{dE_i}{dt} = \frac{1}{2} \sum_j m_j \mathbf{v}_{ij} \cdot \left[\frac{p_i}{\Omega_i \rho_i^2} \nabla_i W_{ij}(h_i) + \frac{p_j}{\Omega_j \rho_j^2} \nabla_i W_{ij}(h_j) + \Pi_{ij} \bullet \nabla_i W_{ij} \right] + \frac{\boldsymbol{\tau}_{\alpha\beta i} \boldsymbol{\epsilon}_{\alpha\beta i}}{\rho_i}$ (7)
Equation of state	$p_i = f(\rho_i, E_i)$ (8)

2.2. Unified SPH formulations

This study explores shockwave propagation and the structural impact of explosions using a unified SPH framework. This approach integrates the fluid conservation equations with the structural constitutive equations.

Table 1 represents the governing equation for SPH formulation. The discretized SPH formulation has various forms of governing equations. In mass conservation, there are two approaches to calculate density in the SPH method, mass summation Eq. (4) and continuity Eq. (5). In this research, the mass summation approach is used. For both momentum and energy conservation, the symmetric form Eq. (6), (7) are chosen and the search range is adjusted to a larger smoothing length to conform with Newton's third law. The equation of state (EOS) is a function of density and internal energy and is different from each material.

The stress tensor $\boldsymbol{\sigma}_{\alpha\beta}$ is defined in terms of the shear stress tensor $\boldsymbol{\tau}_{\alpha\beta}$ and an isotropic tensor $p\delta_{\alpha\beta}$ as follows Eq. (9)

$$\boldsymbol{\sigma}_{\alpha\beta} = -p\delta_{\alpha\beta} + \boldsymbol{\tau}_{\alpha\beta} \quad (9)$$

2.3. Adaptive SPH

The smoothing length h determines the interaction range for each particle. While a constant smoothing length is typically used in conventional SPH methods, certain situations like explosions or impacts may necessitate a variable smoothing length to account for an unstable environment. Consequently, the value of the smoothing length is decided based on variations in density, leading to distinct smoothing lengths for individual particles. An increase in relative density leads to a decrease in smoothing length, and conversely, a decrease in relative density results in an increase in smoothing length [12]. The procedure for calculating the smoothing length incorporates the smoothing length correction factor term Eq. (10). This term is implemented to adjust for fluctuations of properties when the smoothing length is changed in the governing equations

$$\Omega_i = 1 - \frac{\partial h_i}{\partial \rho_i} \sum_j m_j \frac{\rho_{0j}}{\rho_{0i}} \frac{\partial W_{ij}(h_i)}{\partial h} \quad (10)$$

To prevent unphysical oscillation, the unified SPH method contains artificial viscosity. It restrains unphysical particle penetration and minimizes computational inaccuracies in the vicinity of the shock wave front. The formula for artificial viscosity is provided as Eq. (11) [13].

$$\Pi_{ij} = \begin{cases} \frac{-\alpha c_{ij} \mathcal{O}_{ij} + \beta \mathcal{O}_{ij}^2}{\rho_{ij}}, & \mathbf{v}_{ij} \bullet \mathbf{x}_{ij} < 0; \\ 0, & \mathbf{v}_{ij} \bullet \mathbf{x}_{ij} \geq 0; \end{cases} \quad (11)$$

Where, $\mathcal{O}_{ij} = \frac{h_{ij} \mathbf{v}_{ij} \bullet \mathbf{x}_{ij}}{|\mathbf{x}_{ij}|^2 + \varphi^2}$; $c_{ij} = \frac{1}{2}(c_i + c_j)$; $\rho_{ij} = \frac{1}{2}(\rho_i + \rho_j)$; $h_{ij} = \frac{1}{2}(h_i + h_j)$;

$\nabla_i W_{ij} = \frac{1}{2} \{ \nabla_i W_{ij}(h_i) + \nabla_i W_{ij}(h_j) \}$; c is the speed of sound; α, β are artificial viscosity coefficients and $\varphi = 0.1h_{ij}$ is included to prevent numerical instability when two particles are overlapping.

2.4. Elasto-viscoplastic constitutive model for SPH

The elasto-viscoplastic constitutive models are critical in simulating the behavior of materials under extreme conditions, such as those encountered during a steam explosion in a nuclear reactor cavity. These models help predict the response of materials to high strain rates and temperatures. The Jaumann stress rate is commonly utilized for this application Eq. (12).

$$\frac{d\boldsymbol{\tau}_{\alpha\beta}}{dt} = 2G \left(\dot{\boldsymbol{\epsilon}}_{\alpha\beta} - \frac{1}{3} \delta_{\alpha\beta} \dot{\boldsymbol{\epsilon}}_{\gamma\gamma} \right) + \boldsymbol{\tau}_{\alpha\gamma} \dot{\mathbf{R}}_{\beta\gamma} + \boldsymbol{\tau}_{\gamma\beta} \dot{\mathbf{R}}_{\alpha\gamma} \quad (12)$$

Where G is the shear modulus; $\dot{\boldsymbol{\epsilon}}_{\alpha\beta}$ is the strain rate tensor and $\dot{\mathbf{R}}_{\alpha\beta}$ is the rotation rate tensor. Both the strain rate tensor and the rotation rate tensor, along with their discretized SPH formulations, are defined as follows:

$$\dot{\boldsymbol{\epsilon}}_{\alpha\beta} = \frac{1}{2} \left(\frac{\partial v_\alpha}{\partial x_\beta} + \frac{\partial v_\beta}{\partial x_\alpha} \right) \quad (13)$$

$$\dot{\mathbf{R}}_{\alpha\beta} = \frac{1}{2} \left(\frac{\partial v_\alpha}{\partial x_\beta} - \frac{\partial v_\beta}{\partial x_\alpha} \right) \quad (14)$$

and

$$\dot{\boldsymbol{\epsilon}}_{\alpha\beta i} = \frac{1}{2} \sum_j \frac{m_j}{\rho_j} \left[\mathbf{v}_{\alpha j i} \frac{\partial W_{ij}(h_i)}{\partial x_{\beta i}} + \mathbf{v}_{\beta j i} \frac{\partial W_{ij}(h_i)}{\partial x_{\alpha i}} \right] \quad (15)$$

$$\dot{\mathbf{R}}_{\alpha\beta i} = \frac{1}{2} \sum_j \frac{m_j}{\rho_j} \left[\mathbf{v}_{\alpha j i} \frac{\partial W_{ij}(h_i)}{\partial x_{\beta i}} - \mathbf{v}_{\beta j i} \frac{\partial W_{ij}(h_i)}{\partial x_{\alpha i}} \right] \quad (16)$$

The second invariant of the shear stress J_2 is defined as follows Eq. (17)

$$J_2 = \frac{1}{2} \boldsymbol{\tau}_{\alpha\beta} : \boldsymbol{\tau}_{\alpha\beta} \quad (17)$$

The von Mises stress, σ_{vm} is calculated as $\sqrt{3J_2}$. The plastic state is determined by comparing the σ_{vm} with the yield stress σ_Y . When σ_{vm} surpasses σ_Y , the material particle is deemed to be in the plastic state, and the shear stress is consequently adjusted back to the yield surface.

$$\boldsymbol{\tau}_{\alpha\beta} = \begin{cases} \frac{\sigma_Y}{\sigma_{vm}} \boldsymbol{\tau}_{\alpha\beta}, & \sigma_{vm} > \sigma_Y; \\ \boldsymbol{\tau}_{\alpha\beta}, & \sigma_{vm} \leq \sigma_Y. \end{cases} \quad (18)$$

The Johnson-Cook model determines the yield stress σ_Y . This model, represented by Eq. (19), accounts for strain hardening, high strain rates, and the thermal softening characteristic of the material [14].

$$\sigma_Y = (A + B(\epsilon_p)^n) \left(1 + C \ln \frac{\dot{\epsilon}_p}{\dot{\epsilon}_0} \right) (1 - T^{*m}) \quad (19)$$

Where A, B, C, n, m are constants; E_p is the effective plastic strain; $\dot{E}_p$ is the effective plastic strain rate with $\dot{\epsilon}_0$; and T^* is the dimensionless temperature, which is defined as follows:

$$T^* = \frac{T - T_{room}}{T_{melt} - T_{room}} \quad (20)$$

Plastic strain is updated when the particle exists in plastic phase. The increment in plastic strain is defined according to Eq. (21).

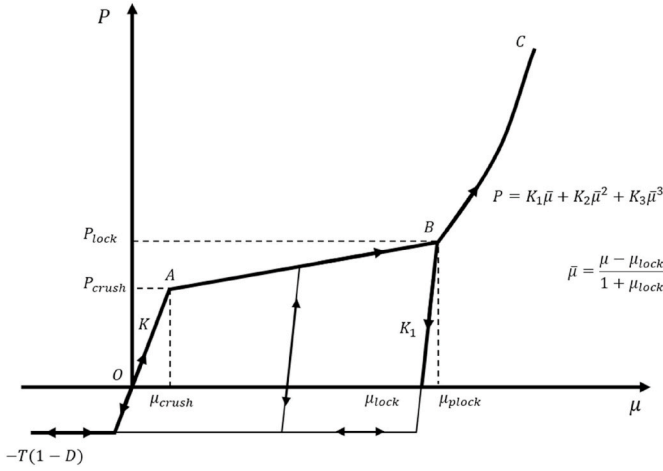
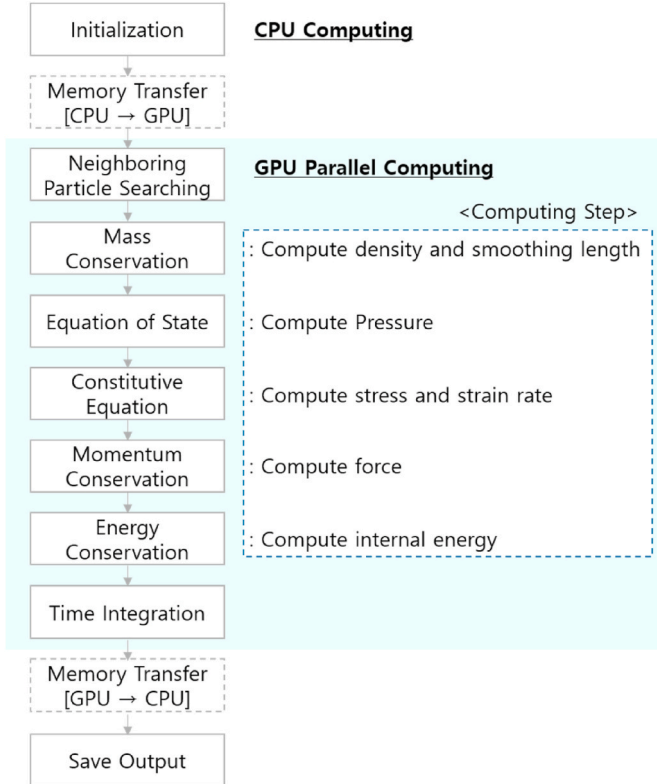
$$\Delta \epsilon_p = \frac{\sigma_{vm} - \sigma_Y}{3G} \quad (21)$$

For concrete material, Holmquist Johnson Cook (HJC) yield model is selected [15]. The HJC model accounts for high-velocity impact loads,

Table 2

Parameter of the Mie-Gruneisen model for water.

Parameter	Water
Initial density ρ_0 (kg/ m ³)	1000
Initial speed of sound C_0 (m/ s)	1480
Coefficient S_1	2.56
Coefficient S_2	1.986
Coefficient S_3	1.2286
Coefficient γ	0.5
Coefficient α	0
Coefficient E_0 (J)	0

**Fig. 2.** Holmquist Johnson Cook equation of state.**Fig. 3.** Unified SPH algorithm.

like those generated by explosions, and is primarily used to describe material behavior under compressive loading.

$$\sigma^* = \begin{cases} [A(1-D) + BP^{*N}](1 + C \ln \epsilon^*) \leq S_{max}, & P^* \geq 0 \\ \left[\frac{AP^*}{T^*(1-D)} + A \right], & P^* < 0 \end{cases} \quad (22)$$

Where $\sigma^* = \sigma_Y/f_c$, f_c is the compressive strength of concrete, S_{max} is the normalized maximum strength that the concrete material can withstand. $P^* = P/f_c$ is the normalized pressure. A, B, N, C are constant. The accumulated damage D is determined by

$$D = \sum \frac{\Delta \epsilon_p + \Delta \mu_p}{\epsilon_p^f + \mu_p^f} \quad (23)$$

Where $\Delta \epsilon_p$ and $\Delta \mu_p$ are the effective plastic strain increment and plastic volumetric strain during a cycle of integration, respectively. $\epsilon_p^f + \mu_p^f$ is the total plastic strain under a constant pressure until fracture, determined by follows Eq. (24).

$$\epsilon_p^f + \mu_p^f = D_1(P^* + T^*)^{D_2} \geq \text{EFMIN} \quad (24)$$

Where $T^* = T/f_c$ is the normalized tensile strength and T is the tensile strength. D_1 , D_2 , EFMIN are constants.

2.5. Equation of state (EOS) model

The Equation of State (EOS) is fundamental in relating thermodynamic quantities within a system, predicting changes in material properties such as density and internal energy. These predictions are crucial for understanding detonation and impact dynamics, as well as modeling material behavior under extreme loading conditions. Various forms of EOS are used depending on the specific requirements of the study. The Jones-Wilkins-Lee (JWL) EOS is employed to characterize the pressure resulting from the detonation [16].

$$P = A \left(1 - \frac{\omega \eta}{R_1} \right) e^{-\frac{R_1}{\eta}} + B \left(1 - \frac{\omega \eta}{R_2} \right) e^{-\frac{R_2}{\eta}} + \omega \eta \rho_0 E \quad (25)$$

Where A, B, R_1 , R_2 , and ω represent constant coefficients dependent on the material; η is the ratio of the explosive material's density to the initial density; ρ_0 denotes the initial density of the explosive material; and E refers to the specific internal energy.

The Mie-Gruneisen EOS is employed to characterize the pressure pertaining to solid metal materials.

$$P = \frac{\rho_0 c_0^2 (\eta - 1) \left[\eta - \frac{\gamma}{2} (\eta - 1) \right]}{[\eta - S(\eta - 1)]^2} + \Gamma E \quad (26)$$

Here, ρ represents the density of the material; c_0 stands for the speed of sound in the material; η is the ratio of the density to the initial density; E denotes the specific internal energy; and Γ , S are the constant coefficients dependent on material.

The Mie-Gruneisen EOS for water, which differs slightly from the EOS used for metals, takes on a unique form under high pressure conditions such as steam explosions to more accurately describe the physical properties of water in extreme environments. The Mie-Gruneisen EOS parameters of water are summarized in Table 2.

$$P = \begin{cases} \frac{\rho_0 C^2 \mu \left[1 + \left(1 - \frac{\gamma_0}{2} \right) \mu - \frac{a}{2} \mu^2 \right]}{\left[1 - (S_1 - 1) \mu - S_2 \frac{\mu^2}{\mu + 1} - S_3 \frac{\mu^3}{(\mu + 1)^2} \right]} + (\gamma_0 + a \mu) E & \mu \geq 0 \\ \rho_0 C^2 \mu + (\gamma_0 + a \mu) E & \mu < 0 \end{cases} \quad (27)$$

The Holmquist Johnson Cook (HJC) EOS is utilized to depict the

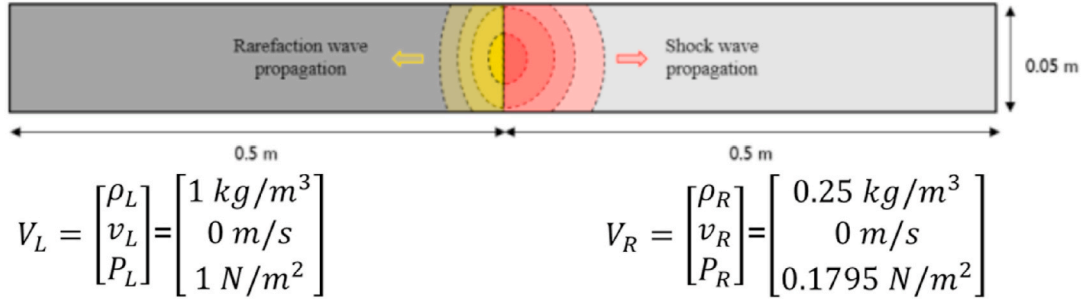
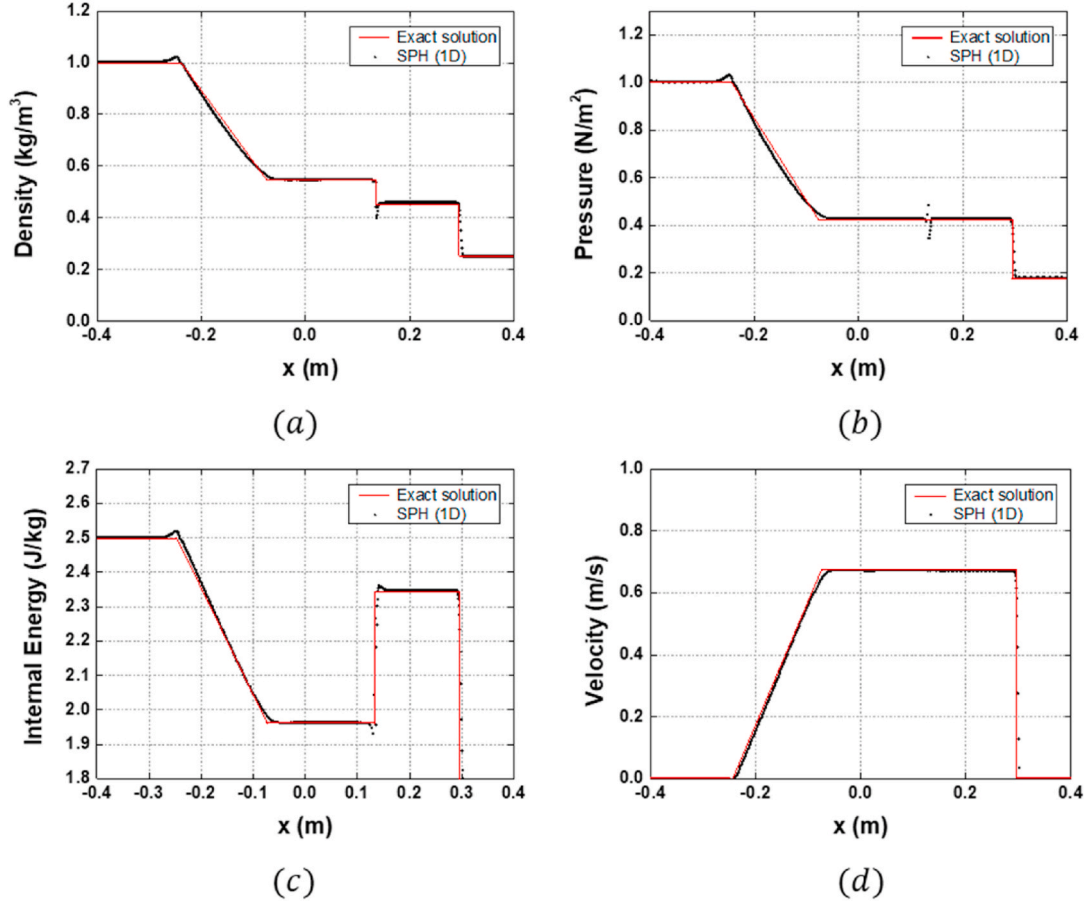


Fig. 4. Geometry of shock tube.

Fig. 5. Shock tube (a) density (b) pressure (c) internal energy (d) x-direction velocity profile at $t = 0.2$ s.

pressure-compaction response for the concrete material. This equation consists of three distinct phases, illustrated in Fig. 2 [17]. The initial phase, referred to as OA, follows a linear elastic response from the negative pressure cutoff $-T(1-D)$ to the elastic limit P_{crush} and is articulated as follows:

$$P = K\mu; \quad P < P_{crush} \quad (28)$$

where the standard volumetric strain is $\mu = \rho/\rho_0 - 1$, $K = P_{crush}/\mu_{crush}$ is the elastic bulk modulus, and μ_{crush} is the volumetric strain at the elastic limit P_{crush} .

The subsequent phase, termed as AB, corresponds to the transitional region. During this phase, air voids are progressively compressed out of the concrete, while plastic volumetric strain induces compaction damage, continuing until the point (μ_{plock}, P_{lock}) . It is expressed as follows Eq. (29):

$$P = \begin{cases} P_{crush} + K_{lock}(\mu - \mu_{crush}) & \text{loading} \\ P_0 - [(1-F)K + FK_1](\mu_0 - \mu) & \text{unloading} \end{cases} \quad (29)$$

where $K_{lock} = (P_{lock} - P_{crush})/(\mu_{plock} - \mu_{crush})$ is the slope, and μ_{plock} is the volumetric strain at the compacting pressure P_{lock} . $F = (\mu_0 - \mu_{crush})/(\mu_{plock} - \mu_{crush})$ is the interpolation factor. P_0 is the pressure when the volumetric strain reached μ_0 prior to unloading.

The final phase, known as BC, represents the compact region where all air voids are considered to be extricated from the concrete. In this phase, the material is assumed to exhibit completely non-linear elasticity, and this relationship is portrayed by Eq. (30):

$$P = \begin{cases} K_1\bar{\mu} + K_2\bar{\mu}^2 + K_3\bar{\mu}^3 & \text{loading} \\ K_1\bar{\mu} & \text{unloading} \end{cases} \quad (30)$$

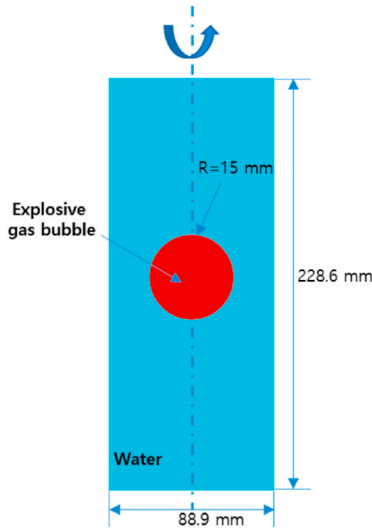
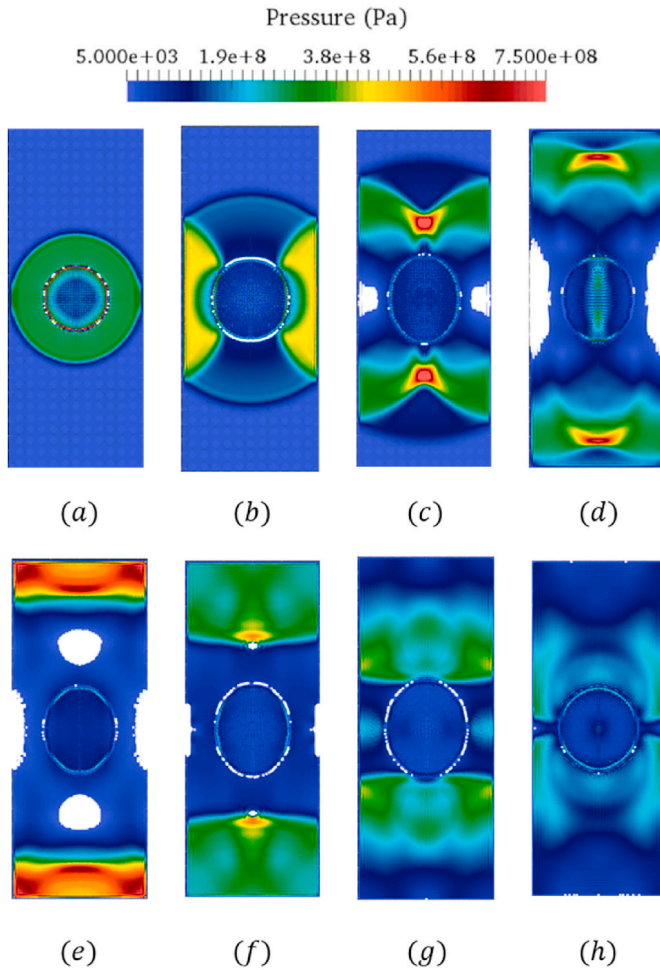


Fig. 6. Initial configuration of UNDEX in rigid tube.

Fig. 7. Pressure contour of UNDEX in rigid tube at different times 15 μ s, 30 μ s, 45 μ s, 60 μ s, 75 μ s, 90 μ s, 105 μ s, and 120 μ s.

where $\bar{\mu} = (\mu - \mu_{lock}) / (1 + \mu_{lock})$ is the modified volumetric strain, and $\mu_{lock} = \rho_{grain} / \rho_0 - 1$ is the locking volumetric strain. ρ_{grain} is the material's grain density, absent of any air voids. K_1 , K_2 , and K_3 are the material constants.

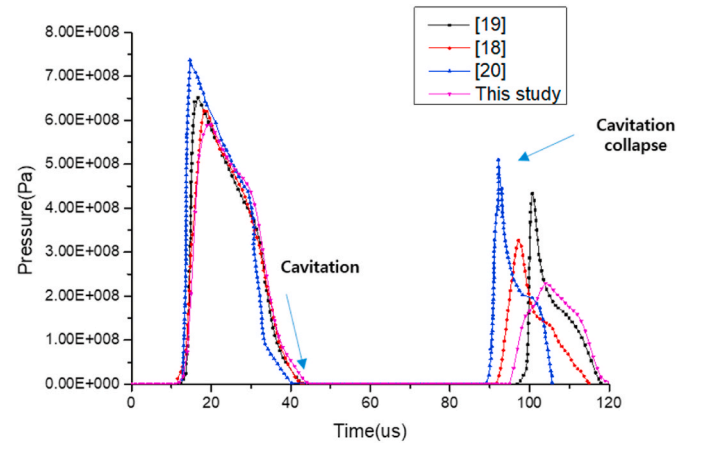


Fig. 8. Pressure profile of UNDEX in rigid tube at mid-line wall.

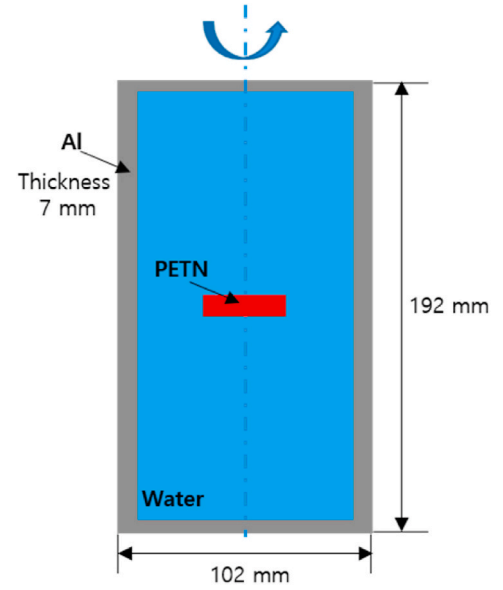


Fig. 9. Initial configuration of UNDEX in deformable tube.

Table 3

Parameter of the Mie-Gruneisen model, Johnson-Cook model and property for Aluminum.

Parameter	Aluminum
Initial density ρ_0 (kg/m ³)	2710
Initial speed of sound c_0 (m/s)	5300
Shear modulus G (GPa)	27.6
Mie-Gruneisen coefficient S	1.5
Mie-Gruneisen coefficient Γ	1.7
Johnson-Cook coefficient A (MPa)	175
Johnson-Cook coefficient B (MPa)	380
Johnson-Cook coefficient C	0.0015
Johnson-Cook coefficient n	0.34
Johnson-Cook coefficient m	1.0
Reference temperature T_{room} (K)	273
Melting point T_{melt} (K)	775

2.6. Algorithm

The following algorithm outlines the steps for conducting unified SPH simulations, which are essential for understanding the dynamic interactions during steam explosions in nuclear reactor cavities.

Table 4
Parameter of JWL EOS model for PETN.

Parameter	PETN
Reference density ρ_0 (kg/ m ³)	1765
Detonation velocity D (m/ s)	8300
Constant A (GPa)	6.17×10^2
Constant B (GPa)	1.6926×10^1
Constant R_1	4.4
Constant R_2	1.2
Constant w	0.25
Detonation energy per unit volume E (J/kg)	5.722×10^6

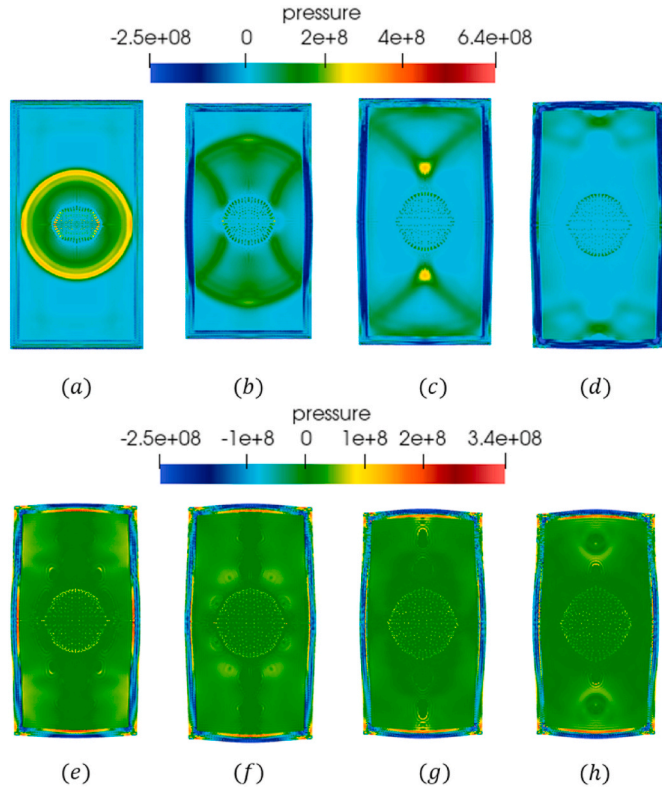
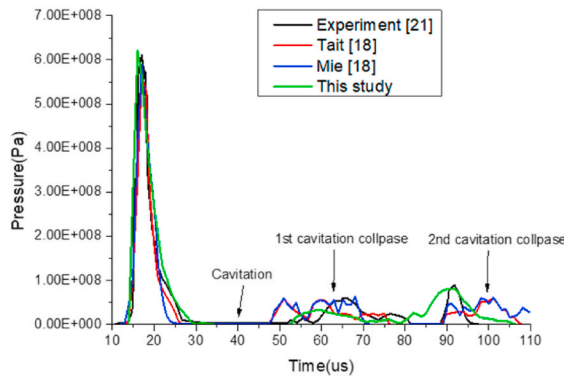
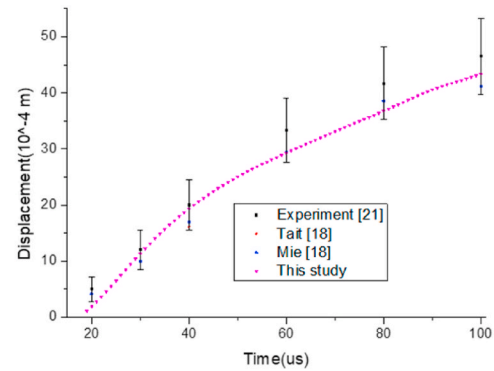


Fig. 10. Pressure contour of UNDEX in deformable tube at different times 15 μ s, 30 μ s, 45 μ s, 60 μ s, 75 μ s, 90 μ s, 105 μ s, and 120 μ s.



(a)



(b)

Fig. 11. (a) Pressure profile and (b) displacement profile of UNDEX in deformable tube at mid-line wall.

As illustrated in Fig. 3, the SPH simulation process begins with the initialization of conditions provided from an input file. Each particle identifies its neighboring particles using the Nearest Neighbor Particle Search (NNPS) method, which is critical for pinpointing the particles that will influence each other's behavior during the simulation. The density of each particle is subsequently calculated based on the mass of the neighboring particles. If necessary, the smoothing length is adjusted to maintain accuracy in the density calculations. Pressure is then calculated using the Equation of State (EOS), a function dependent on the particle's density and internal energy. For particles categorized as solid, a constitutive equation is solved to determine variables such as stress and strain rate, ensuring that material properties are accurately modeled under various conditions. The governing equations are employed to solve the interactions between particles, incorporating parameters like pressure and density. This step effectively models the

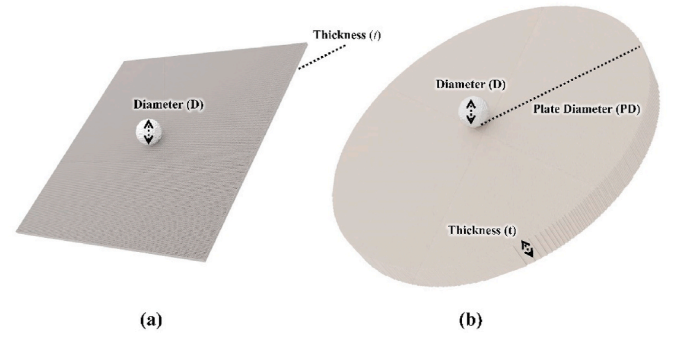


Fig. 12. Initial configuration of high velocity Impact on (a) metal (b) concrete.

Table 5
Parameter of the Mie-Gruneisen model, Johnson-Cook model and property for Copper.

Parameter	Copper
Initial density ρ_0 (kg/m ³)	8930
Initial speed of sound c_0 (m/s)	3940
Shear modulus G (GPa)	45
Mie-Gruneisen coefficient S	1.489
Mie-Gruneisen coefficient Γ	1.98
Johnson-Cook coefficient A (MPa)	90
Johnson-Cook coefficient B (MPa)	292
Johnson-Cook coefficient C	0.02
Johnson-Cook coefficient n	0.31
Johnson-Cook coefficient m	1.09
Reference temperature T_{room} (K)	293
Melting point T_{melt} (K)	1360

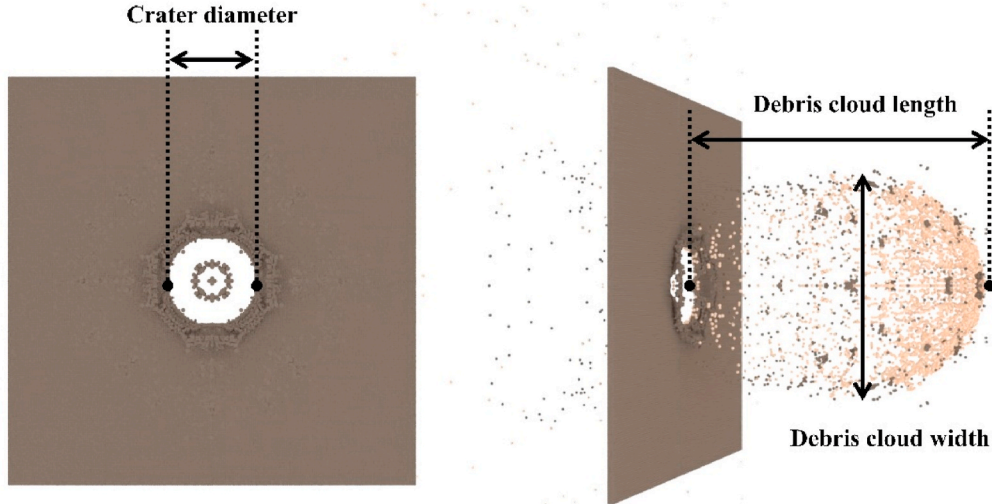


Fig. 13. Representative parameter of HVI simulations.

Table 6
Result of HVI on aluminum plate.

	Crater Diameter (cm)	Debris cloud aspect ratio	Error
Experiment [22]	3.1 (2.75–3.45)	1.39	
Simulation [22]	3.5	1.11	12.9 %, 20.1 %
This study	2.8	1.328 ($\frac{9.3 \text{ cm}}{7 \text{ cm}}$)	9.7 %, 4.5 %

Table 7
Result of HVI on copper plate.

	Crater Diameter (cm)	Debris cloud aspect ratio	Error
Experiment [22]	2.12	1.39	
Simulation [22]	2.3	1.33	8.5 %, 4.3 %
This study	2.1	1.389 ($\frac{7.92 \text{ cm}}{5.7 \text{ cm}}$)	0.94 %, 0.07 %

physical forces and motions within the simulation. The motion of each particle is then calculated for the current time step based on the computed variables, including updated positions and velocities. This iterative process continues for each time step until the predetermined end time of the simulation is reached, ensuring that the dynamic behavior is accurately captured over time. A significant enhancement to the computational speed is achieved through the integration of parallel GPU operations within these calculations. This allows for efficient handling of large-scale simulations, considerably reducing the time required for completion. Fig. 3 provides a visual representation of the described algorithm, encapsulating the initialization, neighbor search, density and pressure calculations, constitutive equation for solids, governing equations, motion calculation, and the iterative process involved in SPH simulations.

3. Verification & validation

The verification and validation processes for the SPH simulation model are presented, focusing on several key scenarios representative of the conditions encountered during a steam explosion in a nuclear

reactor cavity. These scenarios include both one-dimensional (1-D) shockwaves, underwater explosions in both rigid and deformable tubes, and high-velocity impacts on metal and concrete targets. Each of these cases demonstrates the model's capability to accurately simulate various physical phenomena and validate its predictive power against established experimental results or analytical solutions.

3.1. 1-D shockwave

A shock wave is a disturbance that travels faster than the local speed of sound, causing abrupt changes in properties like pressure, density, and velocity. These changes create a discontinuity at the shock front. To verify the SPH model, the Riemann problem was used. This problem, which involved an initial discontinuity in physical variables, is a standard test for hyperbolic partial differential equations. It helps assess how the discontinuity evolves over time, forming shock waves, rarefaction waves, and contact discontinuities. In this test, the SPH model was set up with conditions from a Riemann problem. The simulation tracked the shockwave's evolution, and results were compared with the analytical solution of the Riemann problem.

As illustrated in Fig. 4, the shock tube's initial conditions were established. The tube, filled with gas, was divided by a wall into two sections: one with high pressure (1 Pa) and high density (1 kg/m³), and the other with low pressure (0.1795 Pa) and low density (0.25 kg/m³). When the wall was swiftly removed, a shock wave, a contact discontinuity, and a rarefaction wave were generated. The shock wave traveled into the low-density region, the rarefaction wave moves into the high-density region, and the contact discontinuity follows behind the shock wave. This simulation used about 1000 SPH particles. Initial particle space was 1.25 mm; initial smoothing length was 1.5 times of initial particle space; artificial coefficient $\alpha = 1.0$, $\beta = 10.0$; time step interval was 1×10^{-8} sec.

The analytical solution and the SPH results for density, pressure, internal energy, and x-direction velocity were compared at $t = 0.2$ s. As shown in Fig. 5(a–d), the shock wave reached $x = 0.3$ m, the contact discontinuity appeared at $x = 0.135$ m, and the rarefaction wave reached $x = -0.24$ m. The SPH results accurately depicted the changes in density, pressure, internal energy, and velocity due to the propagation of shock waves.

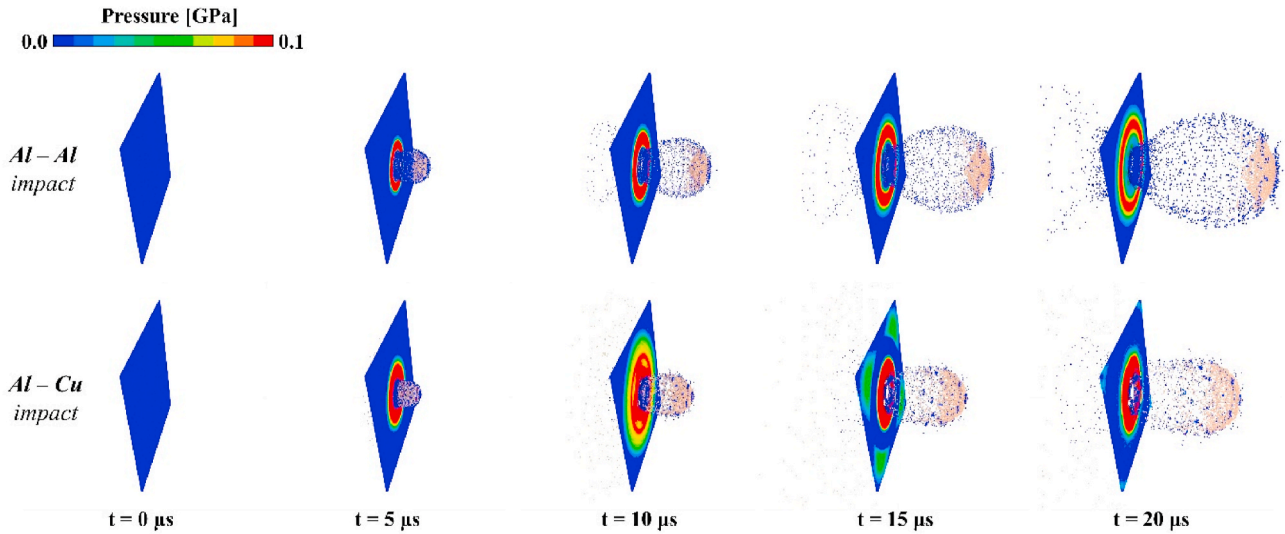


Fig. 14. Dynamic response of the debris cloud from the projectile and target in HVI simulation.

Table 8

Parameter of the Holmquist Johnson Cook (HJC) model for 150 MPa concrete material.

Parameter	Concrete
Compressive strength f_c (MPa)	150
Initial density ρ_0 (kg/m ³)	2700
Shear modulus G (GPa)	20.16
Normalized cohesive strength A	0.79
Normalized pressure hardening B	1.6
Strain rate coefficient C	0.007
Pressure hardening exponent N	0.61
Maximum tensile hydrostatic pressure T (MPa)	7.59
Normalized maximum strength S_{max}	7
Crushing pressure P_{crush} (MPa)	16
Crushing Volumetric strain U_{crush}	0.001
Locking pressure P_{lock} (MPa)	8×10^2
Locking volumetric strain U_{lock}	0.1
Damage constant D_1	0.04
Damage constant D_2	1
Pressure constant K_1 (MPa)	8.5×10^4
Pressure constant K_2 (MPa)	-17.1×10^4
Pressure constant K_3 (MPa)	20.8×10^4

3.2. Underwater explosion in rigid tube

Underwater explosions (UNDEX) involve the detonation of explosives in a water environment, generating unique and complex physical phenomena. Upon detonation, an explosive charge rapidly releases energy, creating a high-pressure shock wave that propagates efficiently through water, allowing it to travel great distances with minimal attenuation compared to air. The initial shock wave exerts intense pressure on nearby objects, potentially causing immediate and severe damage. Additionally, bubble pulses can generate cyclic pressure loads, leading to fatigue and structural failure over time. The intense pressure changes associated with underwater explosions can cause cavitation, where vapor bubbles form and collapse, creating high-pressure microjets that can erode materials and surfaces. As the shock wave propagates through water, its energy diminishes with distance due to geometric spreading, absorption, and reflection off surfaces. Similar phenomena occur in steam explosions, making it essential to verify the SPH model's capability to simulate these events accurately. Therefore, SPH was used to examine these effects and validate the model against the complex interactions and material responses involved in underwater explosions.

Initially, UNDEX in rigid tube case was performed by setting the physical properties of the wall to be the same as those of water. The

initial configuration was set as shown in Fig. 6. The diameter of the tube is 88.9 mm and the height is 288.6 mm. The spherical explosive material has a diameter of 30 mm and is placed in the center of the tube. This explosive material used the same physical properties and EOS as [18]. This simulation used about 1,600,000 SPH particles. Initial particle space was 1 mm; initial smoothing length was 1.2 times of initial particle space; artificial coefficient $\alpha = 1.0$, $\beta = 10.0$; time step interval was 1×10^{-8} sec.

As the high pressure gas exploded, a compressed shock wave was generated. Between 15 and 20 μ s, the shock wave from the explosive gas reached the wall as depicted in Fig. 7(a). The shock wave reflected off the wall and interacted with the explosive gas, generating a rarefaction wave as illustrated in Fig. 7(b). When the rarefaction wave reached the wall, cavitation occurred near the mid-line wall as demonstrated in Fig. 7(c–d). The continuous shock wave caused the cavitation to recompress, initiating the collapse of the cavitation as portrayed in Fig. 7(e–f). Subsequently, the pressure rose again in the collapsed area before decreasing once more due to the shock wave as shown in Fig. 7(g–h).

The pressure profile at the mid-line wall was compared with the other studies [18–20] as shown in Fig. 8. The pressure profile at the mid-line wall was compared with other studies. Key comparison points included the time and value of the peak pressure caused by the first shock wave reaching the wall, the timing of cavitation generation and collapse, and the time and value of the peak pressure resulting from cavitation collapse. Compared to other studies, the time and value of the first peak pressure and the cavitation generation time were similar. However, there were minor differences in the timing of cavitation collapse and the resulting peak pressures, although these differences were observed across all studies.

3.3. Underwater explosion in deformable tube

Subsequently, the results of experiment [21] conducted on aluminum tubes were compared. The experiment was carried out in an aluminum container, allowing for the comparison of pressure and displacement distribution. The initial configuration was set as shown in Fig. 9. The diameter of the tube is 102 mm and the height is 192 mm. The thickness of the aluminum wall is 7 mm. The Mie-Gruneisen EOS and Johnson-Cook parameters for Aluminum are summarized in Table 3. 3g of PETN is placed in the center of the tube. The JWLEOS parameters of PETN are summarized in Table 4. This simulation used about 1,570,000 SPH particles. Initial particle space was 1 mm; initial smoothing length was 1.2 times of initial particle space; artificial coefficient $\alpha = 1.0$, $\beta =$

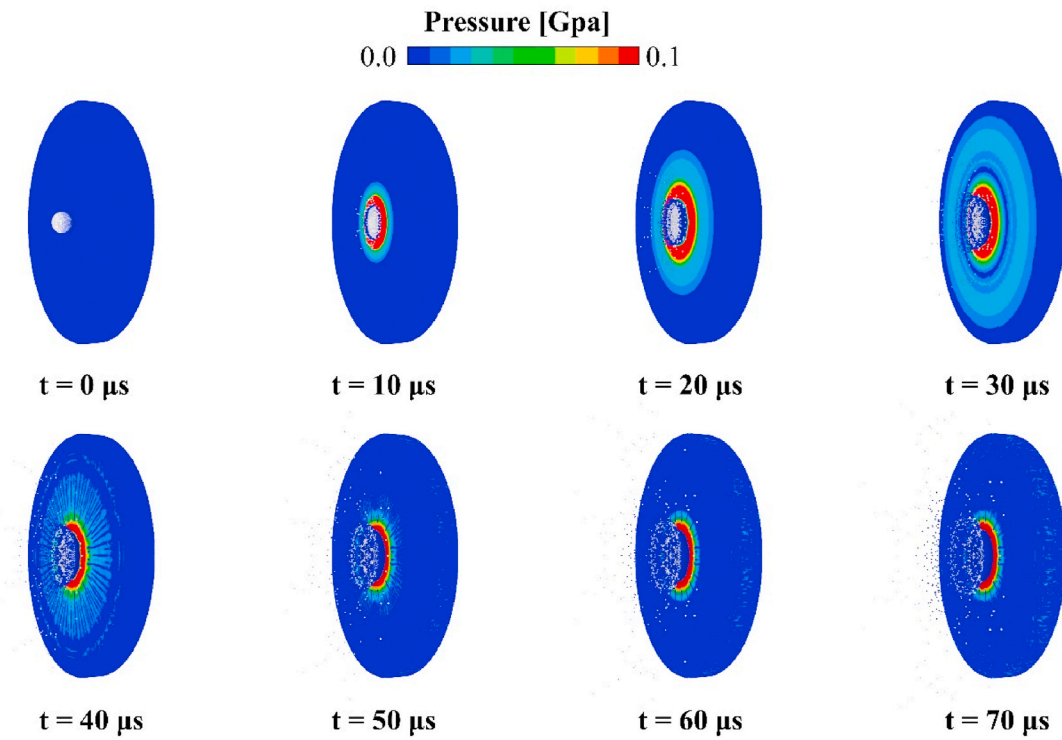


Fig. 15. Pressure distribution of the HVI on concrete at different times time instants with a 10 μ s interval.

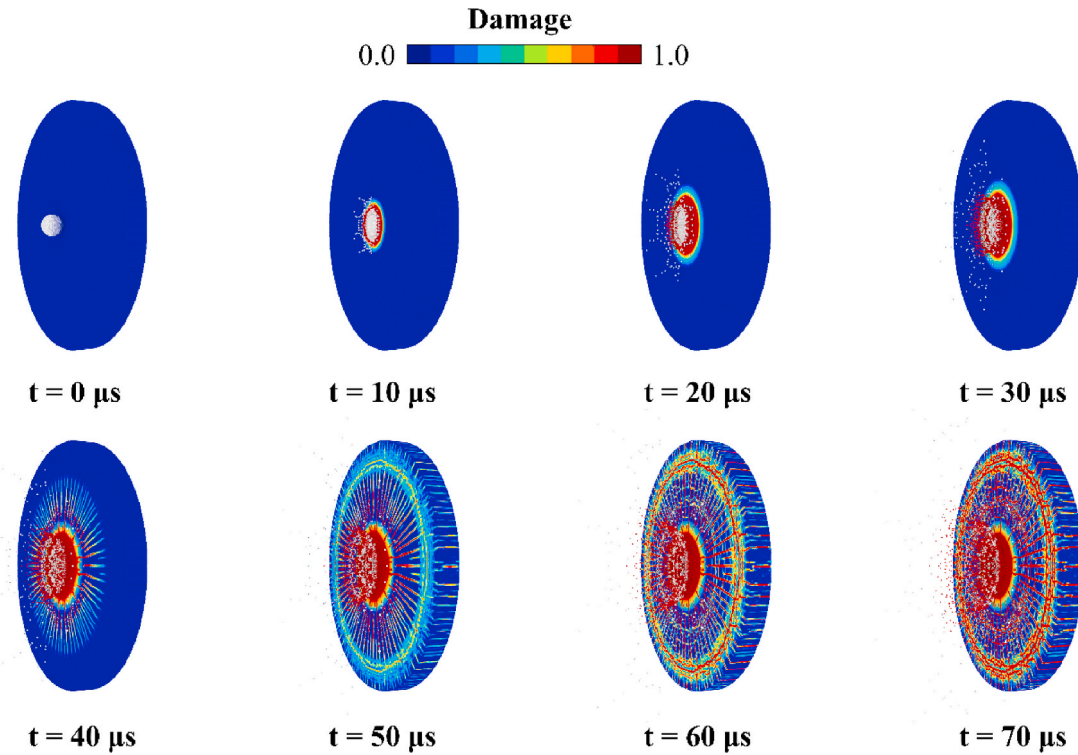


Fig. 16. Damage distribution of the HVI on concrete at different times instants with a 10 μ s interval.

10.0; time step interval was 1×10^{-8} sec.

As shown in Fig. 10, due to the deformation of the wall, the overall pressure value was lower, and the pressure distribution pattern differed slightly compared to the rigid tube simulation. In the latter part of the simulation, the maximum pressure was low, necessitating the use of a lower range to visualize the pressure contour.

The pressure profile, shown in Fig. 11(a), and displacement at the mid-line wall, shown in Fig. 11(b), were compared with experimental results [21] and other research results [18]. The peak pressure caused by the shock wave was found to be similar to that observed in other research. The timing of cavitation generation and the first cavitation collapse aligned better with the experimental results compared to other

Table 9
Result of HVI with concrete plate.

	Crater Diameter (mm)	Error
Experiment [24]	80	
Simulation [23]	85	6.25 %
This study	80	<0.1 %

research. The second cavitation collapse occurred earlier than in other research. Although there were slight differences in the displacement profile at the beginning, the results eventually converged with those of other studies, exhibiting minimal error. The SPH results accurately depict the effects of underwater explosions and the resulting structural loading, validating the model's capability to simulate these complex interactions.

3.4. High velocity impact (HVI) on metal

High velocity impact (HVI) is the collision phenomenon between two objects at extremely high velocities, typically above 1 km/s. When two

objects collide at these velocities, they release enormous amounts of energy, causing significant damage to both objects. During an HVI event, the impact can create a shock wave that can cause further damage to the impacted objects or their surroundings. This shock wave can also generate high temperatures and pressures, leading to material melting, vaporization, and even plasma formation. These conditions result in large deformations, making the simulation difficult for traditional grid-based numerical methods.

To verify the capability of the SPH method to scrutinize phenomena involving large structural loads on the reactor cavity rebar of reinforced concrete, analyses of HVI on metal targets were initially conducted.

Two simulation scenarios were carried out as depicted in Fig. 12(a). The parameters of the Mie-Gruneisen EOS model, Johnson-Cook constitutive model, and properties for Copper are presented in Table 5. Both simulations were performed under the same conditions as the experiments and simulations detailed by Ref. [22], with the results compared. In these simulations, an aluminum ball with a diameter of 1 cm was launched at a thin plate. In one scenario, the ball impacted an aluminum plate of 0.4 cm thickness at a speed of 6.18 km/s. In the other scenario, it impacted a copper plate of 0.15 cm thickness at a speed of

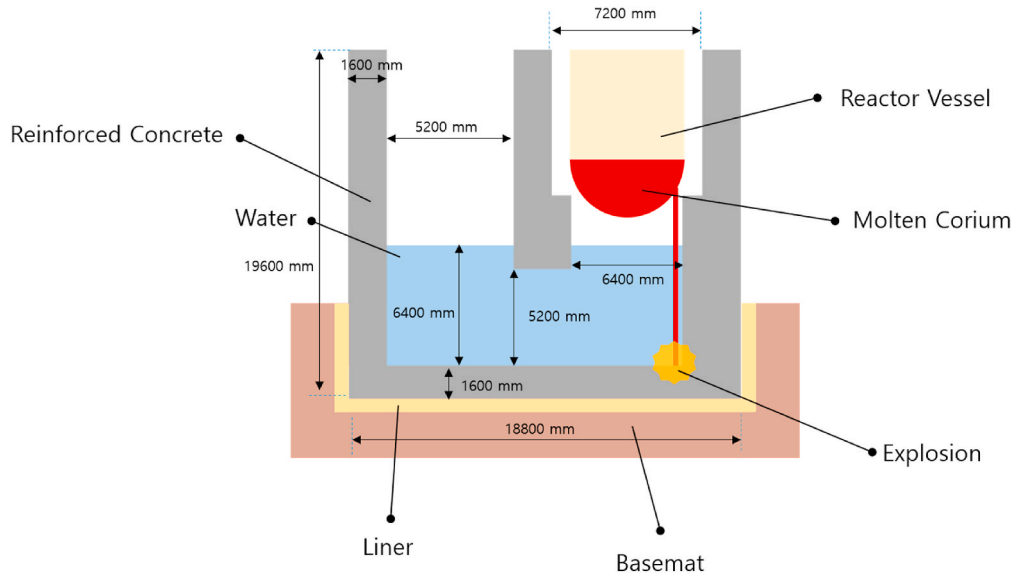


Fig. 17. Schematic of the shockwave propagation in reactor cavity simulation setup.

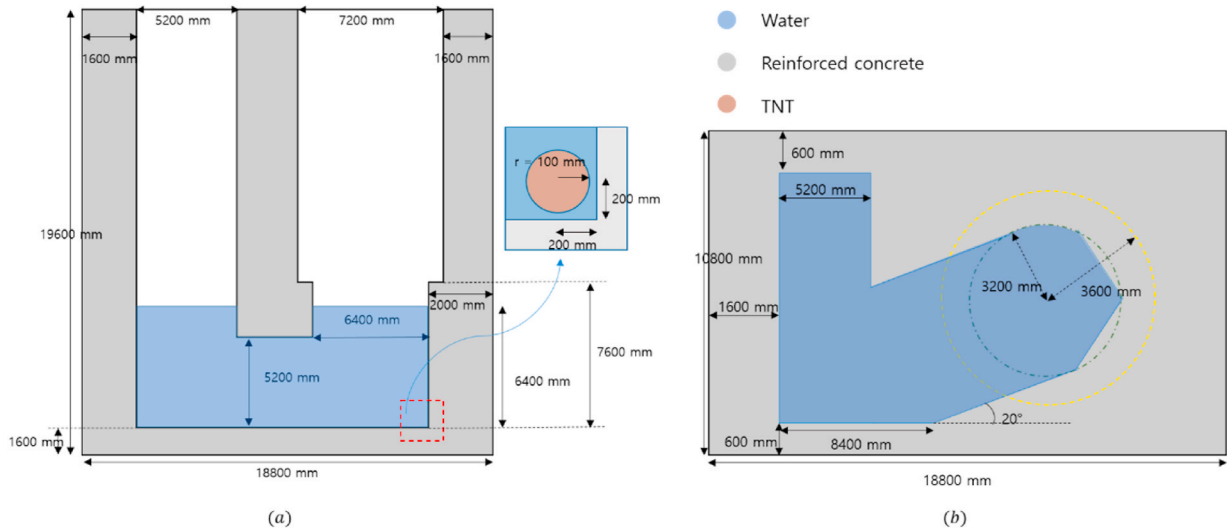


Fig. 18. Specification of reactor cavity (a) cross sectional view (b) top view.

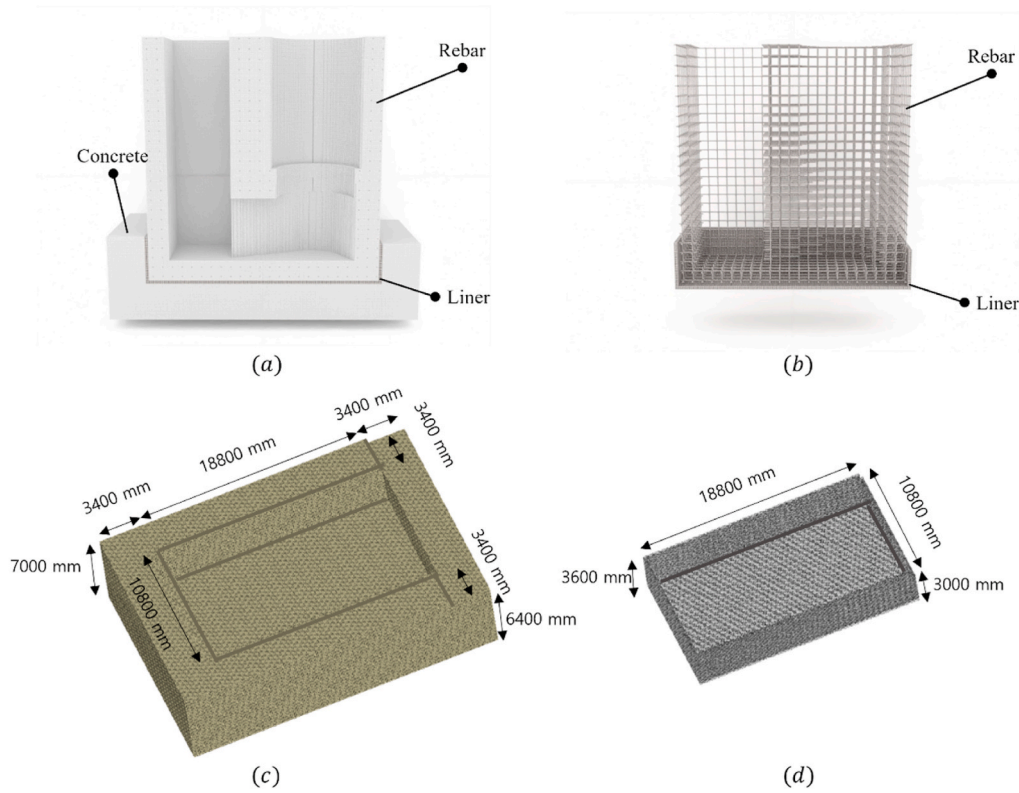


Fig. 19. SPH modeling of the (a) whole component, (b) steel of component, (c) basemat, (d) liner plate.

Table 10

Parameter of the Holmquist Johnson Cook (HJC) model for 44 MPa concrete material.

Parameter	Concrete
Compressive strength f_c (MPa)	44
Initial density ρ_0 (kg/m ³)	2400
Shear modulus G (GPa)	14.86
Normalized cohesive strength A	0.79
Normalized pressure hardening B	1.6
Strain rate coefficient C	0.007
Pressure hardening exponent N	0.61
Maximum tensile hydrostatic pressure T (MPa)	4
Normalized maximum strength S_{max}	7
Crushing pressure P_{crush} (MPa)	14.7
Crushing Volumetric strain U_{crush}	7.4×10^{-4}
Locking pressure P_{lock} (MPa)	8×10^2
Locking volumetric strain U_{lock}	0.1
Damage constant D_1	0.04
Damage constant D_2	1
Pressure constant K_1 (MPa)	1.7×10^4
Pressure constant K_2 (MPa)	3.8×10^4
Pressure constant K_3 (MPa)	2.98×10^4

5.75 km/s. These simulations used about 450,000 and 120,000 SPH particles, respectively. Initial particle space was 0.5 mm; initial smoothing length was 1.2 times of initial particle space; artificial coefficient $\alpha = 1.5$, $\beta = 1.5$; time step interval was 1×10^{-8} sec.

The debris clouds' aspect ratios which defined as the length of the debris cloud divided by its width and crater diameters were evaluated as defined in Fig. 13. An aspect ratio of 1 indicates a perfectly circular shape, while values exceeding 1 represent more elongated debris clouds. Given that the collision energy—determined by the mass and velocity of the impacting object—remains constant across both simulations and experiments, the lower aspect ratios observed in simulations suggest a

Table 11

Parameter of the Mie-Gruneisen model, Johnson-Cook model and property for Weldox 460 E.

Parameter	Weldox 460 E
Initial density ρ_0 (kg/m ³)	7850
Initial speed of sound c_0 (m/s)	5047.6
Shear modulus G (GPa)	78.9
Mie-Gruneisen coefficient S	1.5
Mie-Gruneisen coefficient Γ	2.0
Johnson-Cook coefficient A (MPa)	490
Johnson-Cook coefficient B (MPa)	807
Johnson-Cook coefficient C	0.0114
Johnson-Cook coefficient n	0.73
Johnson-Cook coefficient m	0.94
Reference temperature T_{room} (K)	293
Melting point T_{melt} (K)	1800

reduced conversion of collision energy into thermal energy compared to actual events. The SPH simulations exhibited less error with the experiment than the previous simulation results [22] as shown in Tables 6 and 7. The accuracy of our method was largely due to the careful adjustment of the support domain, ensuring it encompassed enough particles to enhance the precision of particle information calculation. Fig. 14 depicts the time evolution of the debris cloud for both cases. Simulations on plates composed of different materials, aluminum and copper, revealed distinct dynamic behaviors. The more spherical debris clouds observed in aluminum were likely due to its lower density, shear modulus, and melting point compared to copper.

3.5. High velocity impact (HVI) on concrete

A brittle material such as concrete is more prone to suffer severe damage because it tends to fracture and break under stress rather than deform. While ductile materials can undergo significant plastic

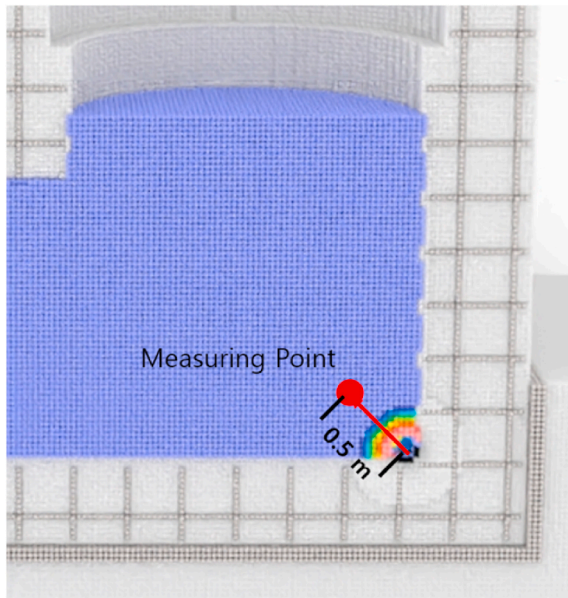


Fig. 20. Measuring point of pressure in reactor cavity.

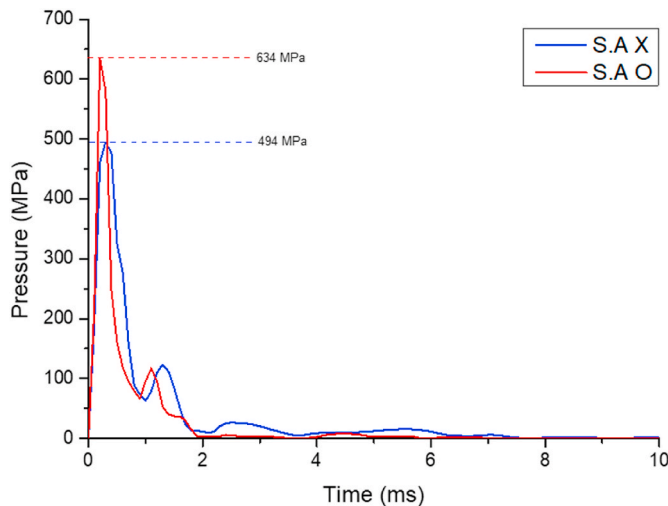


Fig. 21. Pressure profile at the measuring point with and without structural analysis.

deformation, absorbing more energy before failing, brittle materials concentrate stress at the point of impact, leading to rapid crack propagation and failure with little deformation. This difference means that ductile materials are better at absorbing energy and resisting damage, while brittle materials are more susceptible to cracking and breaking under stress. To verify the capability of the SPH method in analyzing phenomena involving large structural loads on the reactor cavity concrete, analysis of HVI on concrete target were conducted.

A simulation scenario similar to the previous metal HVI simulation was conducted, as shown Fig. 12(b). This simulation was carried out under the same conditions as the simulation by Ref. [23] and the corresponding experiment, and the results were compared. In this scenario, a 2.5 cm diameter aluminum ball was launched at a speed of 2 km/s towards a concrete plate with a compressive strength of 150 MPa, a diameter of 30 cm, and a thickness of 2.4 cm, using the HJC model parameters provided in Table 8. This simulation used about 1,700,000 SPH particles. Initial particle space was 1 mm; initial smoothing length was 1.5 times of initial particle space; artificial coefficient $\alpha = 1.5$, $\beta =$

1.5; time step interval was 1×10^{-8} sec.

Figs. 15 and 16 present the numerical predictions of the time-dependent dynamic behavior of a concrete plate and an aluminum projectile during a high-velocity impact (HVI). The figures depict the scene at 10- μ s intervals, showing pressure and damage distribution visually. In Fig. 15, the collision between the aluminum ball and the concrete plate generates a shockwave that travels from the impact point at the center to the edges of the plate. As the shockwave reflects off the edges, the pressure begins to decrease, moving inward from the edges toward the center. This shockwave, followed by the stress wave, initiates and propagates cracks and fractures within the concrete. Initially, 20 radial cracks appear and gradually extend, as shown in Fig. 16. From 50 μ s onward, the reflected stress wave forms a circular crack that becomes visible.

The diameter of the fully damaged area was compared with the results from Ref. [23], as shown in Table 9. The findings confirm that this study successfully replicated the experimental data. A key factor contributing to the accuracy of this study was the inclusion of an artificial viscosity term in Eq. (7). This term was carefully crafted to prevent excessive compression in the energy change rate, enabling the simulation to accurately handle the large deformations of both metal and concrete materials while maintaining adherence to the principle of energy conservation. This validation highlights the effectiveness of our SPH method in addressing the significant deformation challenges associated with metal and concrete materials.

4. Simulation on steam explosion in reactor cavity

4.1. Model description

When a rupture occurs on the side of the reactor vessel molten corium falls into the cavity area below like Fig. 17. Due to the corium falling freely, the initial impact velocity is higher on the side, leading to greater mechanical damage upon impact. As the corium interacts with the residual water present in the cavity, a rapid heat transfer occurs, resulting in the water quickly vaporizing and producing a large volume of steam in a very short time. This rapid generation of steam creates a significant pressure increase, leading to a steam explosion. The force of the explosion generates shockwaves that propagate through the reactor cavity, causing substantial mechanical damage to the containment structures and nearby equipment. Such an event underscores the critical importance of reactor safety systems designed to mitigate the consequences of corium-water interactions, manage shockwave propagation, and contain the effects of any resulting steam explosions.

To investigate the dynamics of a steam explosion within a reactor cavity, a comprehensive computational model based on the SPH method was employed. This model allows for detailed simulation of the fluid-structure interactions and the complex phenomena associated with explosions resulting shockwave propagation. The simulation provides insights into the pressure distribution, mechanical stresses, and potential failure points within the reactor cavity. This approach helps in understanding the critical parameters that influence the severity of the explosion and aids in the development of more effective safety measures and reactor designs to prevent structural failures.

The reactor cavity, as depicted in Fig. 18, features a complex design with specific dimensions and materials, including reinforced concrete, water, and TNT, as shown in both the cross-sectional (Fig. 18(a)) and top views (Fig. 18(b)). This design is fundamentally based on the reactor cavity of the APR1400 [11,25]. The reinforced concrete structure is depicted in gray, forming the main structural components of the cavity. The water, illustrated in blue, fills the lower portion of the cavity, reaching a height of 6400 mm. A cylindrical TNT charge, represented in orange with a radius of 100 mm, is positioned 200 mm from the bottom and 200 mm from the right side of the cavity. The reinforced concrete walls enclose the water-filled area.

The reactor cavity's is modeled using SPH particles. The whole

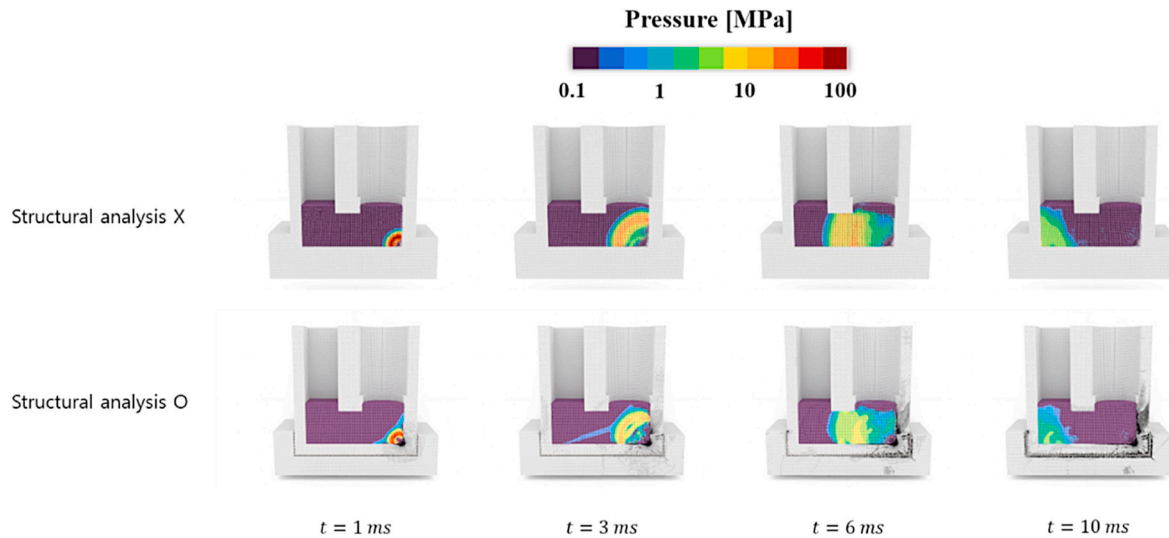


Fig. 22. Cross-sectional pressure contours over time with and without Structural Analysis.

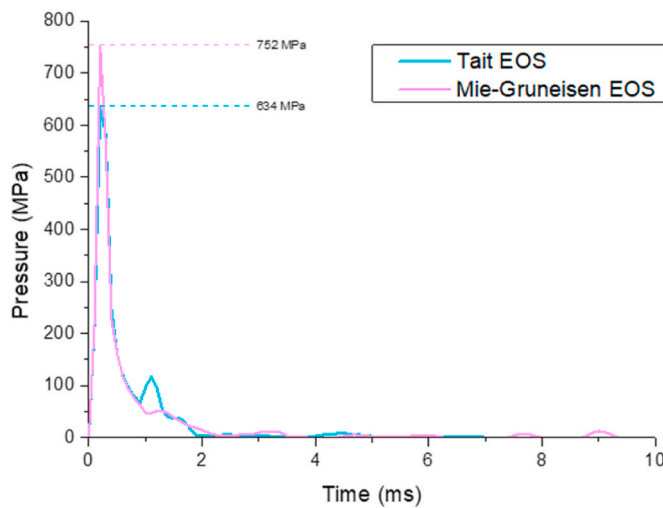


Fig. 23. Pressure profile at the measuring point comparison with different water EOS.

component is modeled as shown in Fig. 19(a), and the steel is arranged at regular intervals within the reactor cavity as shown in Fig. 19(b). This modeling allows for an accurate simulation of the structure's strength and durability. The basemat, which supports the reactor cavity, is shown in Fig. 19(c), and the liner plate between the reactor cavity and the basemat is depicted in Fig. 19(d).

The concrete typically used in reactor cavity and basemat has specified unconfined compressive strengths ranging from 13 to 55 MPa [26]. Therefore, concrete with a compressive strength of 44 MPa was used for this model because it is an appropriate strength within the specified range. The physical properties of the HJC model at 44 MPa are listed in Table 10.

In the APR1400 reactor cavity, the rebar uses A615 Grade 60 steel and the liner uses ASTM A516 Grade 60 steel [8]. Since the coefficients of these two metals are unknown in the material model used in the current simulation, Weldom 460 E steel, which has an elastic modulus and Poisson's ratio similar to those of these two metals, was used in this simulation. The parameters of the Mie-Gruneisen EOS model, Johnson-Cook constitutive model, and properties for Weldom 470 E steel are presented in Table 11.

In order to realize an explosion with the same explosive power as a

steam explosion, the TNT mass equivalent to the explosive power was calculated and placed using TNT equivalence. Assume a scenario where the molten core breaches the side vessel and falls to the bottom of the cavity. For this simulation, the diameter of the molten corium ejected from the reactor pressure vessel is assumed to be 0.1824 m, and the height of the cooling water is about 6.4 m. Consequently, volume of a virtual cylindrical column filled with molten corium is calculated to be 0.16286 m^3 , respectively. With a density of 7960 kg/m^3 for the molten corium [27], the total mass of the corium is determined to be 1331.17 kg. The heat capacity of the molten corium, comprising 80 % fuel (UO_2) and 20 % oxidized cladding material (ZrO_2), is about 1298 kJ/kg at 2800 K [28], resulting in a total thermal energy of around 1,727,855 kJ within the virtual cylindrical column. To determine the equivalent mechanical energy of this thermal energy in terms of TNT, we use an energy conversion ratio of 0.029 % based on measured data ranging from 0.01 % to 0.05 % from the KROTOS experiment with molten corium [29]. This approach is described in detail by Ref. [11]. The mechanical energy equivalent of 1 kg of TNT is approximately 4393 kJ [30]. Consequently, the equivalent TNT mass is calculated to be around 11.41 kg in this study.

This study used about 5,500,000 SPH particles, respectively. Initial particle space was 100 mm; initial smoothing length was 1.5 times of initial particle space; artificial coefficient $\alpha = 1.0$, $\beta = 1.0$; time step interval was 1×10^{-8} sec.

4.2. Shockwave propagation characteristics after explosion

In this section, we analyze the propagation of a shock wave resulting from a steam explosion originating in the lower right corner of the reactor cavity. To achieve this, we examined the cross-sectional pressure contour and the pressure profile at the measuring point shown in Fig. 20. The measuring point is located 0.5 m from the explosion's origin.

We first compare the results with and without structural analysis. In the scenario without structural analysis, the component illustrated in Fig. 19 is modeled as a particle with the same physical properties as water, with its position held fixed.

In the early stages of the explosion (~ 0.3 ms), as illustrated in Fig. 21, the peak pressure reached about 634 MPa when structural analysis was performed, compared to about 494 MPa without structural analysis. This difference, shown in Fig. 22, is caused by the reflection of the shock wave by the structure. However, over time, the shock wave is attenuated due to the presence of the structure, as seen in the cross-sectional pressure contours in Fig. 22. This attenuation occurs because, with structural analysis, the shock wave is both reflected and

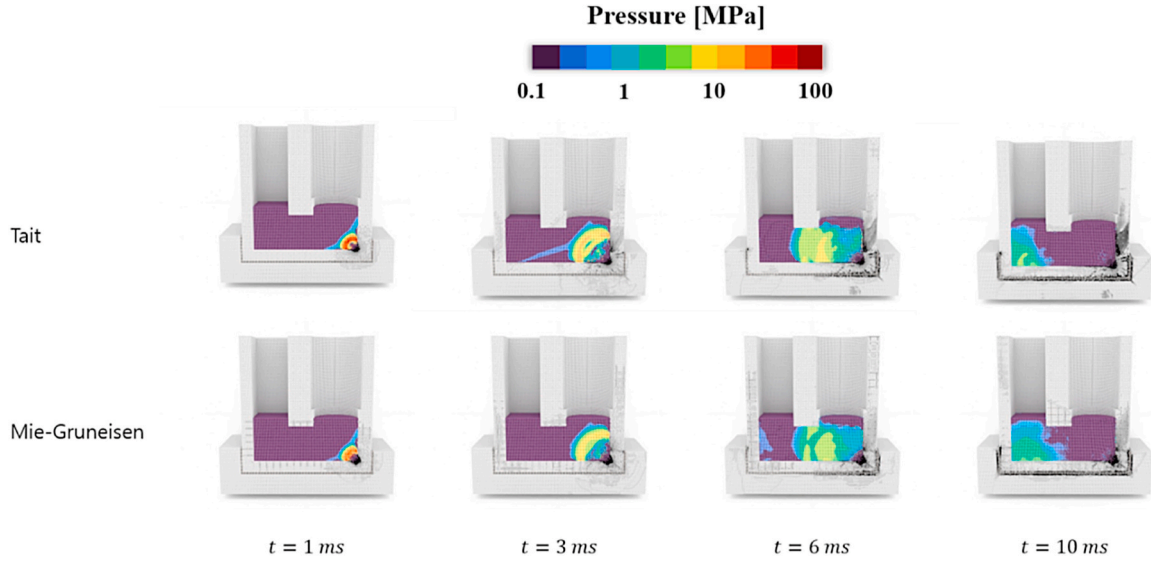


Fig. 24. Cross-sectional pressure contours over time with different water EOS.

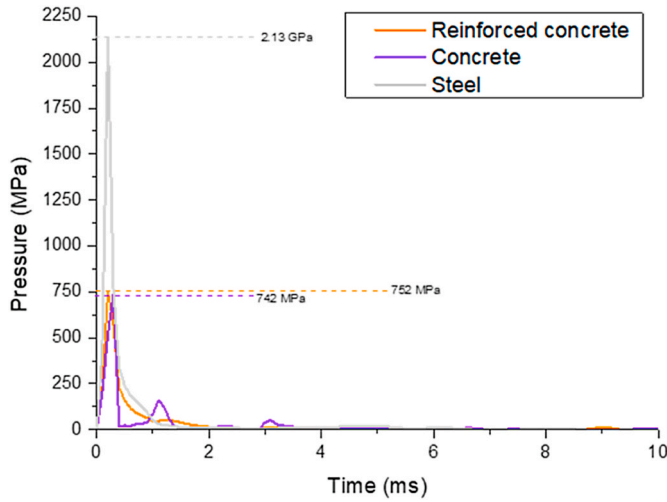


Fig. 25. Pressure profile at the measuring point comparison with different reinforcement ratio.

absorbed by the structure. Initially, the structure reflects most of the shock wave, amplifying it, but as time progresses, the structure absorbs and disperses the shock wave, reducing its intensity. The propagation of the shock wave was found to be faster when the structural analysis was performed. This is due to the higher wave propagation speed within the structure and the complex interactions at the boundary between water and the structure.

Next, the differences in the equation of state (EOS) of water were examined. The comparison was made using the Tait EOS, which is commonly applied for water [18]. According to Ref. [31], the Mie-Grüneisen EOS is a more suitable model for extreme conditions where material is condensed due to high pressure. As shown in Fig. 23, the Mie-Grüneisen EOS results in a peak pressure of approximately 752 MPa in the early stage of the explosion (~ 0.2 ms), which is higher than the peak pressure of about 634 MPa observed with the Tait EOS. However, as time progresses, the difference between the two becomes smaller, and the pressure distributions, depicted in Fig. 24, become somewhat similar. The higher peak pressure observed with the Mie-Grüneisen EOS, which is the primary model used in our study, indicates that it is more conservative from a shock wave analysis perspective.

Next, the analysis results were analyzed according to the difference reinforcement ratio. Generally, when comparing steel and concrete in terms of shock wave behavior, steel tends to reflect shock waves, whereas concrete tends to absorb them. Consequently, as seen in Fig. 25, it can be inferred that a higher reinforcement ratio in reinforced concrete leads to greater shock wave reflection and, as a result, less attenuation. To facilitate a meaningful comparison, the results were examined for three scenarios: the conventional case with a reinforcement ratio of 4.35 %, the case where the reactor cavity is entirely steel (100 % reinforcement ratio), and the case where the reactor cavity is entirely concrete (0 % reinforcement ratio). The analysis results in Fig. 25 indicate that the peak pressure is highest at 2.13 GPa with a 100 % reinforcement ratio, followed by 752 MPa at a 4.35 % reinforcement ratio, and 742 MPa at a 0 % reinforcement ratio. The local peak pressure exhibited minimal differences between pure concrete and reinforced concrete; however, the global pressure distribution displayed a meaningful variance. As seen in Fig. 26, due to the geometry and location of the explosion, the pressure is amplified in the early stage of the explosion by the significant reflection of the shock wave. This amplification is more pronounced when the structure is composed of steel, reflecting the greater extent to which steel amplifies the shock wave compared to concrete.

Finally, the explosive force was increased to observe the differences. A blast force approximately 3.85 times the original was applied at the same location. When the TNT charge was 11.41 kg, the peak pressure was 752 MPa, but when the TNT charge was increased to 44.01 kg, the peak pressure rose to 1.35 GPa, resulting in a peak pressure about 1.79 times greater. Fig. 27 clearly illustrates this comparison, showing the significant increase in peak pressure with the increased explosive force.

The maximum incident pressure in an underwater explosion is represented as Eq. (31) [32], where M is the mass of the explosive charge (kg) and R is the stand off distance (m). Using the conditions provided in Eq. (31), the calculated pressures are approximately 437 MPa for a TNT charge of 11.41 kg and about 726 MPa for a TNT charge of 44.01 kg. These values are lower than the results shown in Fig. 27, likely because Eq. (31) does not account for the reflection by the structure.

$$P_{max} = 53.3 \left(\frac{\sqrt[3]{M}}{R} \right)^{1.13} \quad (31)$$

The peak value of the reflected shock wave pressure can be calculated using a fitting formula, as represented by Eq. (32) [33]. When applying the conditions specified in Eq. (32), the peak reflected shock

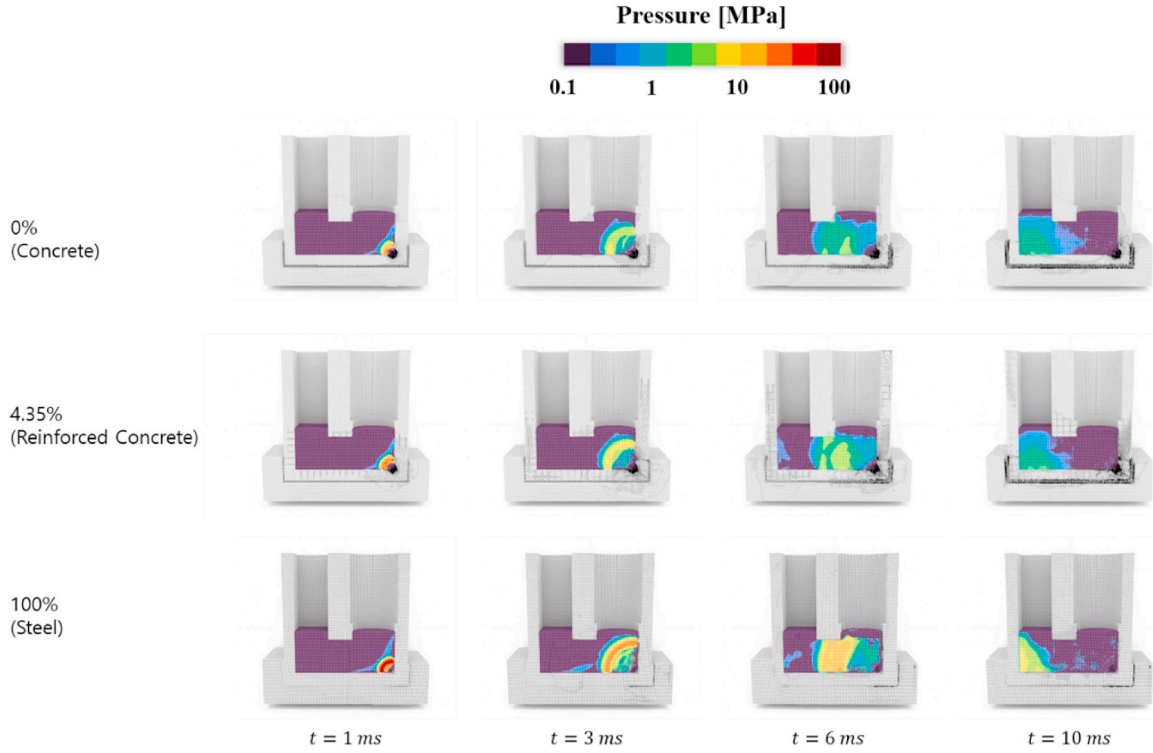


Fig. 26. Cross-sectional pressure contours over time with different reinforcement ratio.

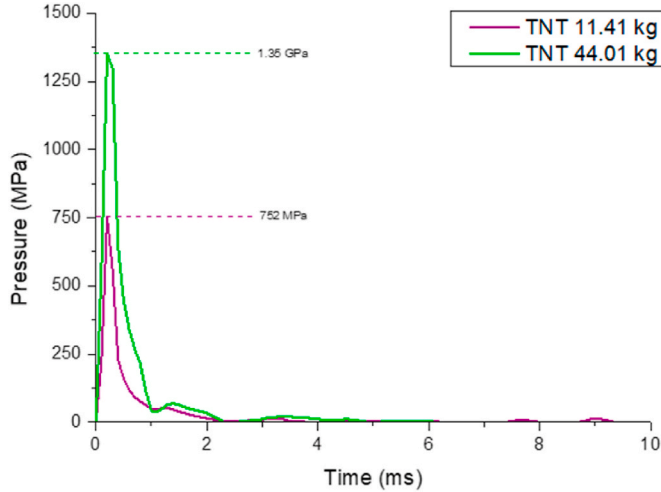


Fig. 27. Pressure profile at the measuring point comparison with different explosive force.

wave pressure is approximately 1.27 GPa for a TNT charge of 11.41 kg and about 2.29 GPa for a TNT charge of 44.01 kg, which is significantly higher than the results shown in Fig. 27. However, this value depends on the structure from which the shock wave reflects. As the explosive charge increases by a factor of 3.85, both the fitting formula and the results closely align, confirming the expected increase in peak pressure.

$$P_{max,r} = 113.3 \left(\frac{\sqrt[3]{m}}{R} \right)^{1.30} \quad (32)$$

Despite this increase, the velocity of the shock wave propagating through the water and the structure did not differ significantly. Fig. 28 demonstrates that while the peak pressure increased, the propagation speed of the shock wave remained relatively consistent across both

scenarios. However, it can also be observed that the distribution of the shock waves changes slightly over time, likely due to their interaction with the structure.

4.3. Impact on concrete containment by shockwave

One method to assess structural integrity is by evaluating whether the maximum principal strain exceeds specific criteria. For steel, which is used in liners and reinforcement, the maximum principal strain should not exceed 0.05 [34], while for concrete, the maximum principal strain should be limited to 0.005 [35]. As the analysis did not exceed these criteria in any case, an additional step was taken to assess the damaged area of the concrete material.

When comparing the different EOSs for water, it was found that the Tait EOS resulted in a larger fully damaged area (damage = 1) compared to the Mie-Grüneisen EOS. Given that the overall pressure distributions of the two EOSs were similar, and the Mie-Grüneisen EOS is known to good at shock wave propagation [31], it can be concluded that the Tait EOS tends to overestimate the shock wave's impact on the structure. Fig. 29 clearly shows this difference in the extent of damage, with the Tait EOS leading to a more extensive area of concrete damage over time.

The comparison of different reinforcement ratios reveals that the 4.35 % reinforcement ratio results in a larger damaged area than the 0 % reinforcement ratio. This is because a higher reinforcement ratio leads to a stronger shock wave, thereby increasing the load on the structure. Fig. 30 clearly illustrates how the extent of damage is greater with the higher reinforcement ratio over time, highlighting the impact of reinforcement ratio on shock wave behavior and structural load.

When comparing the differences in explosive force, it was observed that as the explosive force increases, the load on the structure also increases, leading to a larger damaged area. Fig. 31 clearly shows the progression of damage over time, with a significantly larger damaged area resulting from the higher explosive force of 44.01 kg TNT compared to 11.41 kg TNT.

Additionally, a fully damaged area is observed in the concrete surrounding the steel liner as depicted in Figs. 29–31. This damage is

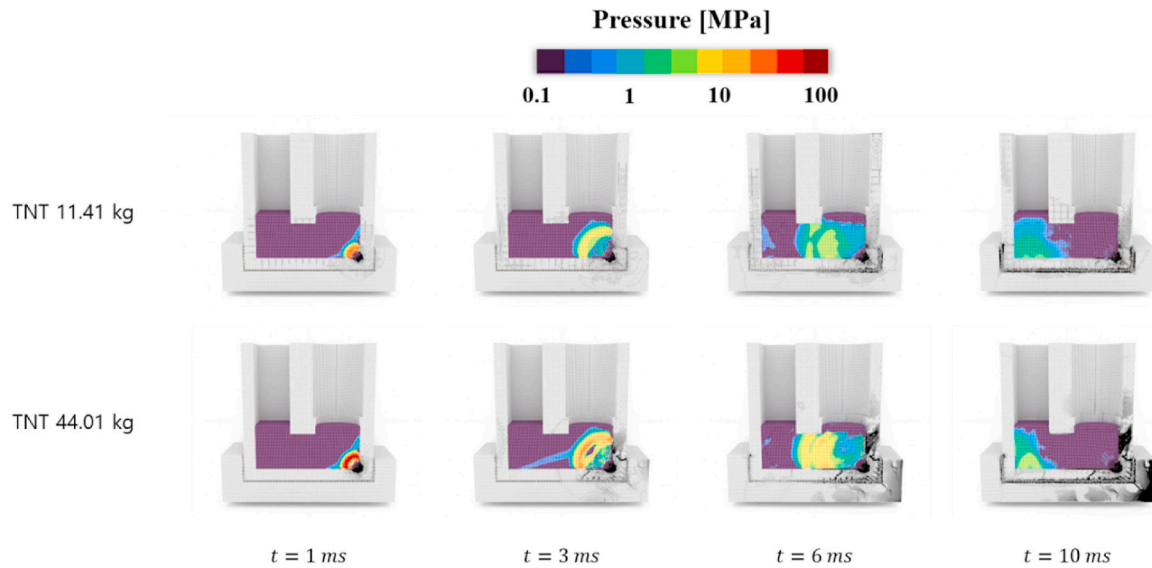


Fig. 28. Cross-sectional pressure contours over time with different explosive force.

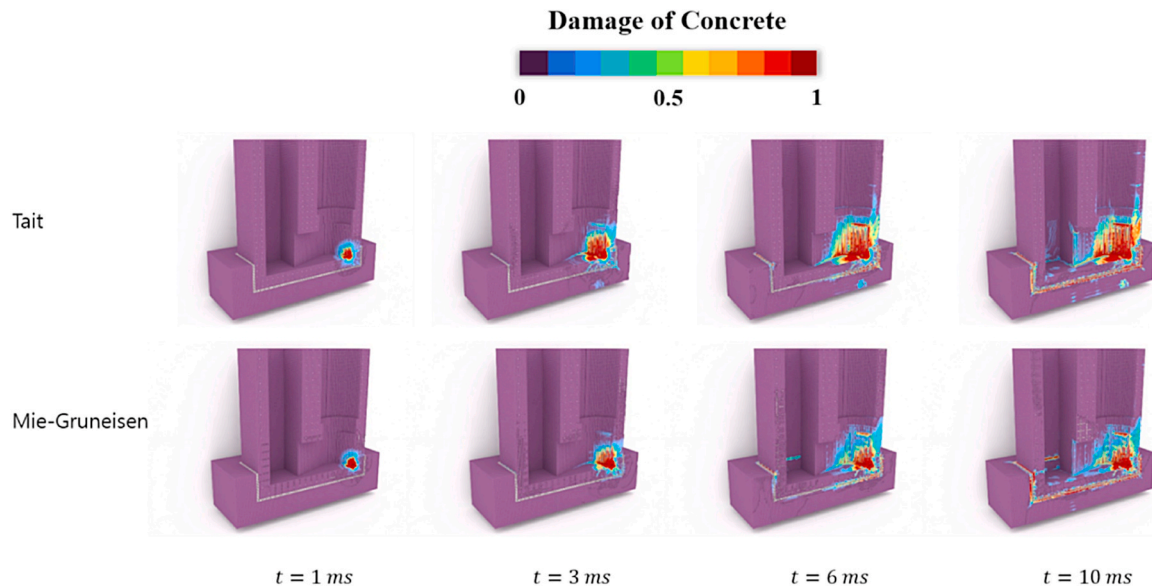


Fig. 29. Cross-sectional damage contours over time with different water EOS.

caused by an impedance mismatch between the two different media, where the significant difference in their acoustic characteristic impedance leads to greater scattering of shock waves at the interface [36]. This scattering, while reducing the overall load on the structure through energy dissipation, also results in increased localized loads around the boundary. Consequently, despite the local damage around the liner, the maximum load on the overall structure is reduced, making the inclusion of a liner beneficial for the integrity of the entire structure.

5. Conclusions

In this study, we introduced a unified SPH method that effectively addresses both shock wave phenomena and structural analysis without requiring integration with other computational methods. Our validations assessed the capabilities of this method through distinct tests, evaluating shock wave analysis and structural analysis independently, followed by complex coupled scenarios. The validity of the SPH method for shock wave analysis was demonstrated using shock tube problems,

while structural analysis capability was affirmed through HVI tests. The method's ability to analyze shock wave-structure interactions was further established through UNDEX tests. All numerical results were consistent with experimental outcomes.

Based on this validation, the analysis simulated the explosive force of a steam explosion caused by a side vessel failure in the APR1400 cavity and examined the propagation of the resulting shock wave and its impact on the structure. In this analysis, the differences between the explosion in the APR1400 cavity with and without structural analysis, the differences between the EOS of the water, the differences among the structural material of the reactor cavity, and the differences in explosion force were examined from the perspective of shock wave propagation and structural impact. The results showed that while the shock wave propagation speed in the water remained consistent across different conditions, the shock wave propagation to the structure, the shock wave intensity, and the degree of structural damage varied significantly under different conditions. However, it was confirmed that in all cases, the maximum principal strain, which serves as the structural integrity

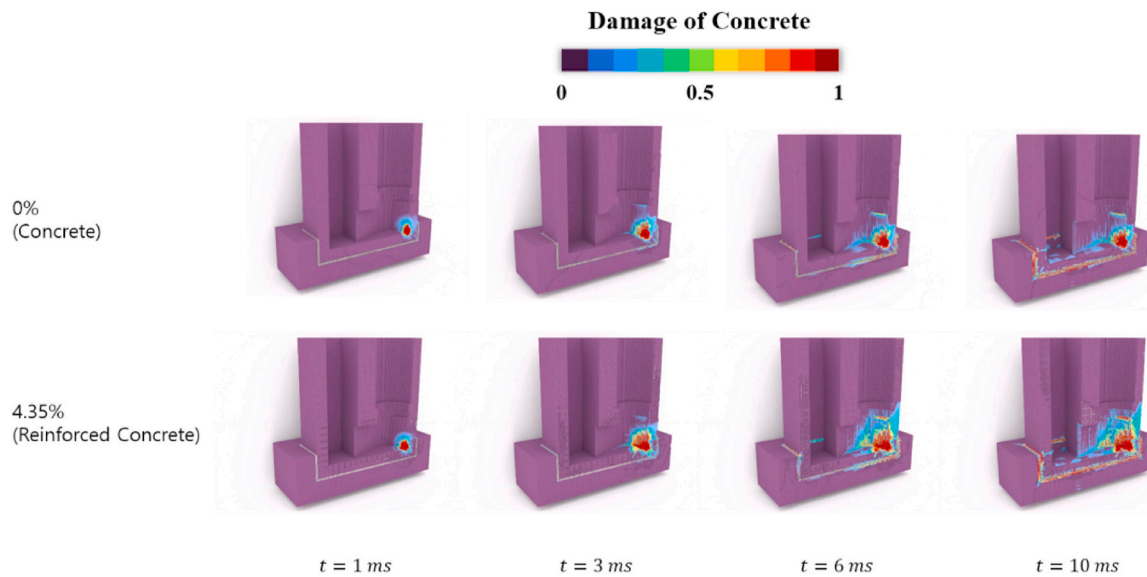


Fig. 30. Cross-sectional damage contours over time with different reinforcement ratio.

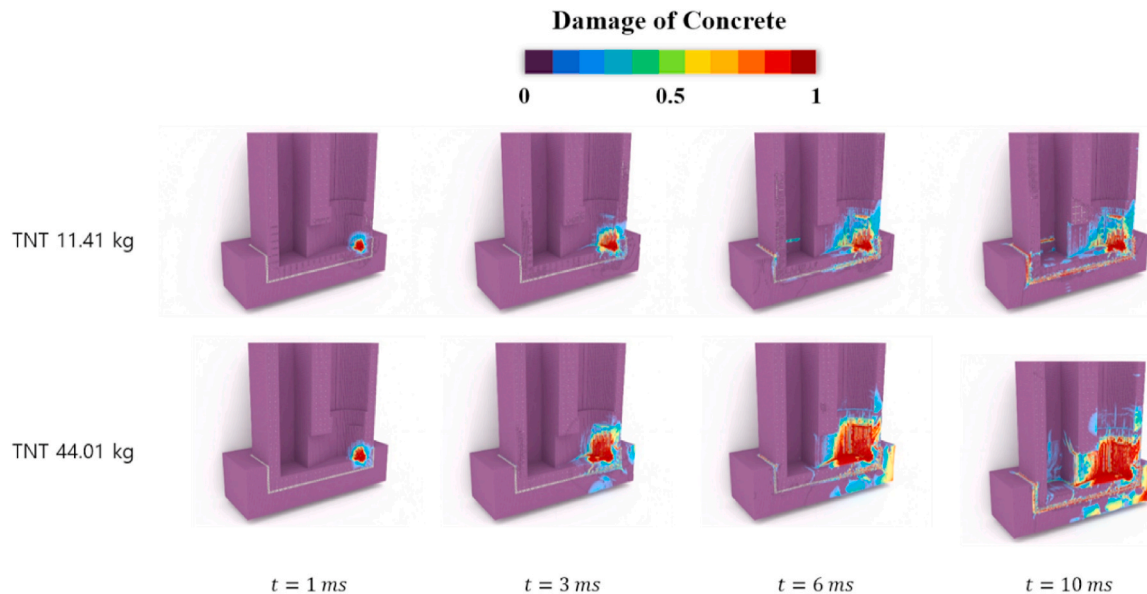


Fig. 31. Cross-sectional damage contours over time with different explosive force.

criteria, did not exceed the criteria, resulting only in localized damage.

The EOS of water was compared using the Tait EOS, which is typically used for water, and the Mie-Grüneisen EOS, which is used under high temperature and pressure conditions. The results showed that the peak pressure was higher for the Mie-Grüneisen EOS, while the overall damaged area of the structure was larger for the Tait EOS. When comparing the materials of the reactor cavity structure—concrete (reinforcement ratio 0 %), steel (reinforcement ratio 100 %), and reinforced concrete (reinforcement ratio 4.35 %)—it was found that the overall pressure was higher in the steel structure than in the other materials. This is because steel has a relatively higher strength than concrete and reflects the shock wave rather than absorbing it. Additionally, the damage area was slightly larger in the reinforced concrete structure compared to the concrete structure. When the explosive force was increased to 3.85 times larger the original explosive force, the peak pressure rose by about 1.8 times. This increase in explosive force also led to a greater load on the structure, resulting in a larger damaged area. These results facilitated comprehensive analyses under various

conditions, enabling qualitative assessments of scenarios likely to yield conservative outcomes, crucial for effective risk management and informed design decisions.

For future works, it would be more accurate to know the specification of the actual APR1400 cavity as the current study used an arbitrary reinforcement ratio and placement due to a lack of available information. Additionally, applying various concrete material models in the analysis could enable a more accurate assessment of the impact on the intensity and propagation of the resulting shock wave, as well as provide a more precise load analysis of the structure according to the concrete material model.

CRediT authorship contribution statement

JinHyun Kim: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Hee Sang Yoo:** Data curation, Formal analysis, Software, Visualization, Writing – review &

editing, Methodology. **Young Beom Jo:** Methodology, Writing – review & editing, Software. **Eung Soo Kim:** Conceptualization, Funding acquisition, Methodology, Project administration, Resources, Software, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Research Foundation of Korea(NRF) grant funded by the Korea government (MSIT). (No. 2021M2D2A1A03046881) This work was also supported by the Nuclear Safety Research Program through the Korea Foundation of Nuclear Safety (KoFONS) using the financial resource granted by the Nuclear Safety and Security Commission (NSSC) of the Republic of Korea (Grant No. 2106033-0222-CG100).

References

- [1] L. Cizelj, et al., Vulnerability of a partially flooded PWR reactor cavity to a steam explosion, *Nucl. Eng. Des.* 236 (2006) 1617–1627.
- [2] M. Leskovar, et al., Ex-vessel steam explosion analysis for pressurized water reactor and boiling water reactor, *Nucl. Eng. Technol.* 48 (2016) 72–86.
- [3] T. Theofanous, et al., "The study of steam explosions in nuclear systems. Advanced Reactor Severe Accident Program.", 1995.
- [4] I. Huhtiniemi, D. Magallon, Insight into steam explosions with corium melts in KROTOS, *Nucl. Eng. Des.* 204 (1–3) (2001) 391–400.
- [5] Jin Ho Song, et al., Insights from the recent steam explosion experiments in TROI, *Journal of nuclear science and technology* 40 (10) (2003) 783–795.
- [6] Seung Hyun Kim, et al., Structural assessment of fully flooded reactor cavity and penetration piping under steam explosion conditions, *Int. J. Pres. Ves. Pip.* 131 (2015) 36–44.
- [7] Seung Hyun Kim, et al., Modeling of reinforced concrete for reactor cavity analysis under energetic steam explosion condition, *Nucl. Eng. Technol.* 48 (1) (2016) 218–227.
- [8] Sang-Hyun Park, Bang Kwang-Hyun, Jong-Rae Cho, Structural integrity evaluation of a reactor cavity during a steam explosion for external reactor vessel cooling, *Energies* 14 (12) (2021) 3605.
- [9] Gregory Carl Besette, *Modeling Coupled Blast/structure Interaction with Zapotec, Benchmark Calculations for the Conventional Weapon Effects Backfill (CONWEB) Tests*. No. SAND2004-4096, Sandia National Laboratories (SNL), Albuquerque, NM, and Livermore, CA (United States), 2004.
- [10] Seung Hyun Kim, Yoon-Suk Chang, Yong-Jin Cho, Parametric analyses of major nuclear components and reinforced concrete structures under FCI-induced explosive condition, *Nucl. Eng. Des.* 322 (2017) 148–158.
- [11] Seong-Kug Ha, Yeo-Hoon Yoon, Numerical simulation of pressure wave propagation and its effect on damage to the reactor cavity under TNT detonation for steam explosion, *Buildings* 13 (9) (2023) 2152.
- [12] Daniel J. Price, et al., Phantom: A Smoothed Particle Hydrodynamics and Magnetohydrodynamics Code for Astrophysics, vol. 35, Publications of the Astronomical Society of Australia, 2018, p. e031.
- [13] M. Monaghan, J. Joe, "Smoothed particle hydrodynamics.", in: *Annual review of astronomy and astrophysics*, 1992, pp. 543–574, 30 (A93-25826 09–90).
- [14] Gordon R. Johnson, A constitutive model and data for metals subjected to large strains, high strain rates and high temperatures. *Proceedings of the 7th International Symposium on Ballistics*, 1983, p. 1983. The Hague, Netherlands.
- [15] Ren, Gen-Mao, et al., "Parameters of Holmquist–Johnson–Cook Model for High-Strength Concrete-like Materials under Projectile impact." *International Journal of Protective Structures* 8.3 (2017): 352–367.
- [16] R. Pramanik, D. Deb, Implementation of smoothed particle hydrodynamics for detonation of explosive with application to rock fragmentation, *Rock Mech. Rock Eng.* 48 (2015) 1683–1698.
- [17] Xiaochao Jin, et al., Ballistic resistance and energy absorption of honeycomb structures filled with reactive powder concrete prisms, *J. Sandw. Struct. Mater.* 19 (5) (2017) 544–571.
- [18] Jian-Yu Chen, et al., Simulations for the explosion in a water-filled tube including cavitation using the SPH method, *Computational Particle Mechanics* 6 (2019) 515–527.
- [19] T.G. Liu, B.C. Khoo, W.F. Xie, Isentropic one-fluid modelling of unsteady cavitating flow, *J. Comput. Phys.* 201 (1) (2004) 80–108.
- [20] Ali Jafarian, Ahmadreza Pishevar, An exact multiphase Riemann solver for compressible cavitating flows, *Int. J. Multiphas. Flow* 88 (2017) 152–166.
- [21] G. Chambers, et al., Pressure measurements on a deforming surface in response to an underwater explosion in a water-filled aluminum tube, *Shock Vib.* 8 (1) (2001) 1–7.
- [22] S. Hiermaier, et al., Computational simulation of the hypervelocity impact of Al-spheres on thin plates of different materials, *Int. J. Impact Eng.* 20 (1–5) (1997) 363–374.
- [23] Jian-Yu Chen, et al., Numerical modeling of the damage mechanism of concrete-soil multilayered medium subjected to underground explosion using the GPU-accelerated SPH, *Eng. Anal. Bound. Elem.* 151 (2023) 265–274.
- [24] Sidney Chocron, et al., Hypervelocity impact on concrete and sandstone: momentum enhancement from tests and hydrocode simulations, in: *Hypervelocity Impact Symposium*, vol. 883556, American Society of Mechanical Engineers, 2019.
- [25] Hanbee Na, et al., Analysis of concrete ablation depth by MCCI assuming top flooding condition of APRI400 NPP, *Ann. Nucl. Energy* 160 (2021) 108363.
- [26] Dan J. Naus, "Primer on Durability of Nuclear Power Plant Reinforced Concrete Structures-A Review of Pertinent factors.", 2007.
- [27] R. Sairanen, et al., OECD Research Programme on Fuel-Coolant Interaction Steam Explosion Resolution for Nuclear Applications-SERENA, Organisation for Economic Co-Operation and Development, 2007. Final Report-December 2006. No. NEA-CSNI-R-2007-11.
- [28] M. Bottin, et al., Molten Corium–Concrete Interaction: investigation of convective heat transfer in a pool with gas sparging, *Ann. Nucl. Energy* 93 (2016) 35–42.
- [29] H.-S. Park, R. Chapman, M.L. Corradini, Vapor explosions in a one-dimensional large-scale geometry with simulant melts. Division of Safety [ie Systems] Analysis and Regulatory Effectiveness, Office of Nuclear Regulatory Research, US Nuclear Regulatory Commission, 1999.
- [30] R. Rajendran, K. Narasimhan, Deformation and fracture behaviour of plate specimens subjected to underwater explosion—a review, *Int. J. Impact Eng.* 32 (12) (2006) 1945–1963.
- [31] Olivier Heuzé, "Complete forms of mie-gruneisen equation of state." *AIP conference proceedings* 1793, AIP Publishing, 2017.
- [32] Robert H. Cole, Weller Royal, Underwater explosions, *Physics Today* 1 (6) (1948) 35, 35.
- [33] Zhuang, Tie-shuan, et al., Experimental investigation on dynamic response and damage models of circular RC columns subjected to underwater explosions, *Defence Technology* 16 (4) (2020) 856–875.
- [34] Nuclear Energy Institute (NEI), "NEI 07-13 [Revision 8P] Methodology for Performing Aircraft Impact Assessments for New Plant Designs.", April 2011.
- [35] IAEA, *Safety Aspects of Nuclear Power Plants in Human Induced External Events: Assessment of Structures; Safety Reports Series. No.87*, International Atomic Energy Agency, Vienna, Austria, 2018.
- [36] Shiming Zhuang, Guruswami Ravichandran, Dennis E. Grady, An experimental investigation of shock wave propagation in periodically layered composites, *Journal of the Mechanics and Physics of Solids* 51 (2) (2003) 245–265.